

Modality-Agnostic Decoding of Vision and Language from fMRI

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eLife Assessment

This manuscript introduces a potentially **valuable** large-scale fMRI dataset pairing vision and language, and employs rigorous decoding analyses to investigate how the brain represents visual, linguistic, and imagined content. The current manuscript blurs the line between a resource paper and a theoretical contribution, and the evidence for truly modality-agnostic representations remains **incomplete** at this stage. Clarifying the conceptual aims and strengthening both the dataset technicality and the quantitative analyses would improve the manuscript's significance for the fields of cognitive neuroscience and multimodal AI.

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Abstract

Humans perform tasks involving the manipulation of inputs regardless of how these signals are perceived by the brain, thanks to representations that are agnostic to the stimulus modality. Investigating such modality-agnostic representations requires experimental datasets with multiple modalities of presentation. In this paper, we introduce and analyze SemReps-8K, a new large-scale fMRI dataset of 6 subjects watching both images and short text descriptions of such images, as well as conditions during which the subjects were imagining visual scenes. The multimodal nature of this dataset enables the development of modality-agnostic decoders, trained to predict which stimulus a subject is seeing, irrespective of the modality in which the stimulus is presented. Further, we performed a searchlight analysis revealing that large areas of the brain contain modality-agnostic representations. Such areas are also particularly suitable for decoding visual scenes from the mental imagery condition. The dataset will be made publicly available.

Introduction

Several regions in the human brain have developed a high degree of specialization for particular lower-level perceptive as well as higher-level cognitive functions (Kanwisher, 2010 ) . For many higher-level functions, it is crucial to be able to manipulate inputs regardless of the modality in which a stimulus was perceived by the brain. Such manipulations can be performed thanks to representations that are abstracted away from particularities of specific modalities, and are

therefore *modality-agnostic*. A range of theories have been developed to explain how and where in the human brain such abstract representations are created (Damasio, 1989 [↗](#); Binder et al., 2009 [↗](#); Martin, 2016 [↗](#); Barsalou, 2016 [↗](#); Ralph et al., 2017 [↗](#)).

In order to study how modality-agnostic information is represented in the brain and to exploit it for modality-agnostic decoding, large multimodal neuroimaging datasets with well-controlled stimuli across modalities are required. While large-scale datasets exist for vision (Huth et al., 2012 [↗](#); Chang et al., 2019 [↗](#); Allen et al., 2022 [↗](#)), language (Brennan and Hale, 2019 [↗](#); Nastase et al., 2021 [↗](#); Schoffelen et al., 2019 [↗](#); Tang et al., 2023a [↗](#)), and video (naturalistic movies) (Aliko et al., 2020 [↗](#); Visconti di Oleggio Castello et al., 2020 [↗](#); Boyle et al., 2020 [↗](#)), none of these contain a controlled set of equivalent stimuli that are presented separately in both modalities. For instance, the different modalities in movies (vision and language) are complementary but do not always carry the same semantics. Further, they are not presented separately but simultaneously, impeding a study of the respective activity pattern caused by each modality in isolation.

Here, we present SemReps-8K, a new large-scale multimodal fMRI dataset of 6 subjects each viewing more than 8,000 stimuli which are presented separately in one of two modalities, as images of visual scenes or as descriptive captions of such images. In addition, the dataset also contains 3 imagery conditions for each subject, where they had to imagine an visual scene based on a caption description they had received before the start of the fMRI experiment. We exploit this new data to develop decoders that are specifically trained to leverage modality-agnostic patterns in the brain. Such modality-agnostic decoders are trained on brain imaging data from multiple modalities, which we demonstrate here for the case of vision and language. In contrast to *modality-specific* decoders that can be applied only in the modality that they were trained on, *modality-agnostic* decoders can be applied to decode stimuli from multiple modalities, even without knowing a priori the modality the stimulus was presented in.

We find that modality-agnostic decoders trained on this dataset perform on par with their respective modality-specific counterparts for decoding images, despite the additional challenge of uncertainty about the stimulus modality. For decoding captions, the modality-agnostic decoders even outperform the respective modality-specific decoders (because the former, but not the latter, can leverage the additional training data from the other image modality).

Additionally, we use this novel kind of decoders for a searchlight analysis to localize regions with modality-agnostic representations in the brain. Previous studies that aimed to localize modality-agnostic patterns were based on limited and rather simple stimulus sets and did not always agree on the exact location and extent of such regions (e.g. Vandenberghe et al., 1996 [↗](#); Shinkareva et al., 2011 [↗](#); Devereux et al., 2013 [↗](#); Fairhall and Caramazza, 2013 [↗](#); Jung et al., 2018 [↗](#)). We design a searchlight analysis based on a combination of modality-agnostic decoders and cross-decoding. The results reveal that modality-agnostic patterns can be found in a widespread left-lateralized network across the brain, encompassing virtually all regions that have been proposed previously.

Finally, we find that modality-agnostic decoders trained only on data with perceptual input also generalize to conditions during which the subjects were performing mental imagery. There is a large overlap in the areas that we identified as modality-agnostic and those that are suitable for decoding mental imagery.

Related Work

Modality-agnostic representations

Decades of neuro-anatomical research (e.g. based on clinical lesions) and electrophysiology in non-human and human primates, as well as modern experiments leveraging recent brain imaging techniques have provided evidence that the activation patterns in certain brain areas are modality-specific; for example, the occipital cortex responds predominantly to visual stimulation (Felleman and Van Essen, 1991 [↗](#); Sereno et al., 1995 [↗](#); Grill-Spector and Malach, 2004 [↗](#)), and a commonly left-lateralized network responds to language processing tasks (Zola-Morgan, 1995 [↗](#); Fedorenko et al., 2010 [↗](#), 2011 [↗](#); Friederici, 2017 [↗](#); Brennan, 2022 [↗](#)).

More recent research has started to focus on higher-level regions that respond with *modality-agnostic* patterns, i.e., patterns that are abstracted away from any modality-specific information. A modality-agnostic region responds with similar patterns to input stimuli of the same meaning, even if they are presented in different modalities (e.g. the word “cat” and picture of a cat). Such regions have also been described as abstract/conceptual (Binder, 2016 [↗](#)), modality-invariant (Man et al., 2012 [↗](#)), modality-independent (Dirani and Pykkänen, 2024 [↗](#)), supramodal (Sanchez et al., 2020 [↗](#)), or amodal (Fairhall and Caramazza, 2013 [↗](#)) (although see distinction made in Barsalou, 2016 [↗](#)).

Several theories and frameworks on how the brain forms modality-agnostic representations from modality-specific inputs have been proposed. The convergence zones view proposes that information coming from modality-specific sensory cortices is integrated in multiple convergence zones that are distributed across the cortex, predominantly in temporal and parietal lobes (Damasio, 1989 [↗](#); Tranel et al., 1997 [↗](#); Meyer and Damasio, 2009 [↗](#)). These convergence zones are organized hierarchically, learned associations are used to create abstractions from lower-level to higher-level feature representations (Simmons and Barsalou, 2003 [↗](#); Meyer and Damasio, 2009 [↗](#)). A perceived stimulus first causes activity in the related low-level modality-specific region (e.g. visual cortex), subsequently higher-level convergence zones serve as relays that cause associated activity in other regions of the brain (e.g. the language network) (Meyer and Damasio, 2009 [↗](#); Kiefer and Pulvermüller, 2012 [↗](#)). According to Binder, 2016 [↗](#), the most high-level convergence zones can become so abstract that they are representing amodal symbols.

The GRAPES framework (Grounding Representations in Action, Perception, and Emotion Systems) also suggests that representations are distributed across temporal and parietal areas of the cortex. More specifically, they are hypothesized to be situated in areas connected to the perception and manipulation of the environment, as well as in the language system (Martin, 2009 [↗](#), 2016 [↗](#)). According to this theory, conceptual knowledge is organized in domains: For example, semantic information related to object form and object motion is represented within specific visual processing systems, regardless of the stimulus modality, and both for perception as well as imagination.

The hub-and-spoke theory states that cross-modal interactions are mediated by a single modality-agnostic hub, located in the anterior temporal lobes (Rogers et al., 2004 [↗](#); Lambon Ralph et al., 2006 [↗](#); Patterson and Lambon Ralph, 2016 [↗](#); Ralph et al., 2017 [↗](#)). The hub contains a “continuous distributed representation space that expresses conceptual similarities among items even though its dimensions are not independently interpretable” (Frisby et al., 2023 [↗](#), p. 262). The spokes form the links between the hubs and the modality-specific association cortices. Most importantly, semantic representations are not solely based in the hub, for a given concept all spokes that are linked to modalities in which the concept can be experienced do contribute to the semantic representation. This explains why selective damage to spokes can cause category-specific deficits (Pobric et al., 2010 [↗](#)).

This is conceptually similar to some aspects of the Global Workspace Theory, which assumes both a multimodal convergence of inputs towards a specific (network of) region(s), and the possibility of flexibly recruiting unimodal regions into this Global Workspace (Baars, 1993 [↗](#), 2005 [↗](#)).

While these and other theories partly disagree on *how* modality-agnostic information is represented in the brain, they agree that such information is distributed across the cortex, and possibly overlapping with the semantic network (Binder et al., 2009 [↗](#); Huth et al., 2012 [↗](#); Andrews et al., 2014 [↗](#); Zwaan, 2016 [↗](#)).

Decoding of vision and language from fMRI

Early approaches of brain decoding focused on identifying and reconstructing limited sets of simple visual stimuli (Haxby et al., 2001 [↗](#); Cox and Savoy, 2003 [↗](#); Kay et al., 2008 [↗](#); Naselaris et al., 2009 [↗](#); Nishimoto et al., 2011 [↗](#)). Soon after, attempts to decode linguistic stimuli could identify single words and short paragraphs with the help of models trained to predict features extracted from word embeddings (Pereira et al., 2018 [↗](#)).

More recently, large-scale open source fMRI datasets for both vision and language have become available (Chang et al., 2019 [↗](#); Allen et al., 2022 [↗](#); Schoffelen et al., 2019 [↗](#); Tang et al., 2023b [↗](#)) and allowed for the training of decoding models for a larger range and more complex naturalistic stimuli with the help of features extracted from deep learning models. For example, modality-specific decoders for vision can be trained by mapping the brain activity of subjects viewing naturalistic images to feature representation spaces of computational models of the same modality (i.e. vision models) (Shen et al., 2019 [↗](#); Beliy et al., 2019 [↗](#); Lin et al., 2022 [↗](#); Takagi and Nishimoto, 2023 [↗](#); Ozcelik and VanRullen, 2023 [↗](#)). Moreover, a range of studies provided evidence that certain representations can transfer between vision and language by evaluating decoders in a modality that they were not trained on (Shinkareva et al., 2011 [↗](#); Man et al., 2012 [↗](#); Fairhall and Caramazza, 2013 [↗](#); Simanova et al., 2014 [↗](#); Jung et al., 2018 [↗](#)). The performance in such *cross-modal decoding* evaluations always lags behind when compared to within-modality decoding. One explanation is that modality-specific decoders are not explicitly encouraged to pick up on modality-agnostic features during training, and modality-specific features do not transfer to other modalities.

To address this limitation, we here propose to directly train *modality-agnostic decoders*, i.e. models that are exposed to multiple stimulus modalities during training in order to make it more likely that they are leveraging representations that are modality-agnostic. Training this kind of decoder is enabled by the multimodal nature of our fMRI dataset: The stimuli are taken from COCO, a multimodal dataset of images with associated descriptive captions (Lin et al., 2014 [↗](#)). During the experiment the subjects are exposed to stimuli in both modalities (images and captions) in separate trials. Crucially, we can map the brain activity of each trial (e.g. the subject viewing an image) to modality-agnostic features extracted from both modalities (the image and the corresponding caption) when training the decoder models. After training, a single modality-agnostic decoder can be used to decode stimuli from multiple modalities, leveraging representations that are common to all modalities.

Decoding of mental imagery

Apart from decoding perceived stimuli, it is also possible to decode representations when subjects were performing mental imagery, without being exposed to any perceptual input. Different theories on mental imagery processes emphasize either the role of the early visual areas (Kosslyn et al., 1999 [↗](#); Pearson, 2019 [↗](#)) or the role of the high-level visual areas in the ventral temporal cortex and frontoparietal networks (Spagna et al., 2021 [↗](#); Hajhajate et al., 2022 [↗](#); Liu et al., 2025 [↗](#)). There is evidence for both kinds of theories in the form of neuroimaging studies that used decoding to identify stimuli during mental imagery. Some of these found relevant patterns in the early visual cortex (Albers et al., 2013 [↗](#); Naselaris et al., 2015 [↗](#)), others highlighted the role of

higher-level areas in the ventral visual processing stream (Stokes et al., 2009; Reddy et al., 2010; Lee et al., 2012; VanRullen and Reddy, 2019; Boccia et al., 2019) as well as the precuneus and the intraparietal sulcus (Johnson and Johnson, 2014). These discrepancies can possibly be explained by differences in experimental design: For example, the early visual cortex might only become involved if the task requires the imagination of high-resolution details, which are represented in lower levels of the visual processing hierarchy (Kosslyn and Thompson, 2003).

Crucially, it has been shown that decoders trained exclusively on trials *with* perceptual input can generalize to imagery trials (Stokes et al., 2009; Reddy et al., 2010; Lee et al., 2012; Johnson and Johnson, 2014; Naselaris et al., 2015), providing evidence that representations formed during perception overlap to some degree with representations formed during mental imagery (Dijkstra et al., 2019).

In our study, we explored to what extent these findings hold true for more varied and complex stimuli. Following previous approaches, we use decoders trained exclusively on trials where subjects were viewing images and captions, and evaluated them on their ability to decode the imagery trials. We additionally hypothesized that mental imagery should be at least as conceptual as it is sensory and therefore primarily recruit modality-agnostic representations. Consequently, modality-agnostic decoders should be ideally suited to decode mental imagery and outperform modality-specific decoders on that task.

Methods for localizing modality-agnostic regions

The first evidence for the existence of modality-agnostic regions came from observations of patients with lesions in particular cortical regions which lead to deficits in the retrieval and use of knowledge across modalities (Warrington and Shallice, 1984; Warrington and McCarthy, 1987; Gainotti, 2000; Damasio et al., 2004). Semantic impairments across modalities have also been observed in patients with the neurodegenerative disorder semantic dementia (Warrington, 1975; Snowden et al., 1989; Jefferies et al., 2009).

In early work exploring the possible locations of modality-agnostic regions in healthy subjects, brain activity was recorded using imaging techniques while they were presented with a range of concepts in two modalities (e.g. words and pictures). Regions that were active during semantic processing of stimuli in the first modality were compared to regions that were active during semantic processing of stimuli in the second modality. The conjunction of these regions was proposed to be modality-agnostic (Vandenberghe et al., 1996; Moore and Price, 1999; Bright et al., 2004).

While this methodology allows for the identification of candidate regions in which semantic processing of multiple modalities occurs, it can not be used to probe the information represented in these regions. In order to compare the information content (i.e. multivariate patterns) of brain regions, researchers have developed Representational Similarity Analysis (RSA, Kriegeskorte et al., 2008) as well as encoding and decoding analyses (Naselaris et al., 2011). More specifically, RSA has been used to find modality-agnostic regions by comparing activation patterns of a candidate region when subjects are viewing stimuli from different modalities (Devereux et al., 2013; Handjaras et al., 2016; Liuzzi et al., 2017). This comparison is performed in an indirect way, by measuring the correlation of dissimilarity matrices of activation patterns. In turn, cross-decoding analysis can be leveraged to identify modality-agnostic regions by training a classifier to predict the category of a stimulus in a given modality, and then evaluating its performance to predict the category of stimuli that were presented in another modality (Shinkareva et al., 2011; Man et al., 2012; Fairhall and Caramazza, 2013; Simanova et al., 2014; Jung et al., 2018). However, all these studies relied on a predefined set of stimulus categories, and can therefore not be easily extended to more realistic and complex stimuli, as we perceive them in our everyday life.

We summarize candidates for modality-agnostic regions that have been identified by previous studies in Appendix 3. This overview reveals substantial disagreement regarding the possible locations of modality-agnostic patterns in the brain. For example, Fairhall and Caramazza, 2013 [↗](#) found modality-agnostic representations in the left ventral temporal cortex (fusiform, parahippocampal, and perirhinal cortex), middle and inferior temporal gyrus, angular gyrus, parts of the prefrontal cortex as well as the precuneus. Shinkareva et al., 2011 [↗](#) found a larger network of left-lateralized regions, including additionally the left superior temporal, inferior parietal, supramarginal, inferior and inferior occipital, precentral and postcentral gyrus, supplementary motor area, intraparietal sulcus, cuneus, posterior cingulum as well as the right fusiform gyrus and the superior parietal gyrus, paracentral lobule on both hemispheres. In contrast, Jung et al., 2018 [↗](#) found modality-agnostic representations only in the right prefrontal cortex. These diverging results can probably be explained by the limited number of stimuli as well as the use of artificially constructed stimuli in certain studies.

Recent advances in machine learning have enabled another generation of fMRI analyses based on large-scale naturalistic datasets. Here, we present a new multimodal dataset of subjects viewing both images and text. Most importantly, the dataset contains a large number of naturalistic stimuli in the form of complex visual scenes and full sentence descriptions of the same type of complex scenes, instead of pictures of single objects and words as commonly used in previous studies. This data enables the development of modality-agnostic decoders that are explicitly trained to leverage features that are shared across modalities. Further, we use this data to localize modality-agnostic regions in the brain by applying decoders in a multimodal searchlight analysis.

Methods

fMRI Experiment

Six subjects (2 female, age between 20 and 50 years, all right-handed and fluent English speakers) participated in the experiment after providing informed consent. The study was performed in accordance with French national ethical regulations (Comité de Protection des Personnes, ID 2019-A01920-57). We collected functional MRI data using a 3T Philips ACHIEVA scanner (gradient echo pulse sequence, TR=2s, TE=30ms, 46 slices with a 32-channel head coil, slice thickness=3mm with 0.2mm gap, in-plane voxel dimensions 3×3mm). At the start of each session, we further acquired high-resolution anatomical images for each subject (voxel size=1mm³, TR=8.13ms, TE=3.74ms, 170 sagittal slices).

Scanning was spanned over 10 sessions (except for sub-01: 11 sessions), each consisting of 13 to 16 runs during which the subjects were presented with 86 stimuli. Each run started and ended with an 8s fixation period. The stimulus type varied randomly inside each run between images and captions. Each stimulus was presented for 2.5 seconds at the center of the screen (visual angle: 14.6 degrees; captions were displayed in white on a dark gray background (font: “Consolas”), the inter-stimulus interval was 1s. Every 10 stimuli there was a fixation trial that lasted for 2.5s. Every 5min there was a longer fixation trial for 16s.

Subjects performed a one-back matching task: They were instructed to press a button when-ever the stimulus matched the immediately preceding one (cf. **Figure 1** [↗](#) Panel A). In case the previous stimulus was of the same modality (e.g. two captions in a row), the subjects were instructed to press a button if the stimuli matched exactly. In the cross-modal case (e.g. an image followed by a caption), the button had to be pressed if the caption was a valid description of the image, and vice versa. Positive one-back trials occurred on average every 10 stimuli.

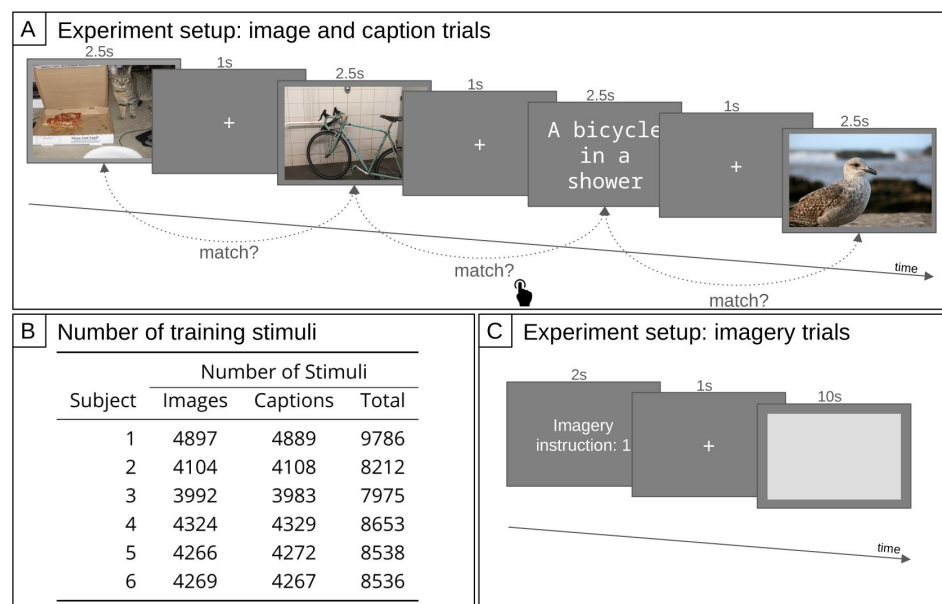


Figure 1.

Panel A: Setup of the main fMRI experiment.

Subjects were seeing images and captions in random alternation. Whenever the current stimulus matched the previous stimulus, the subjects were instructed to press a button (one-back matching task). Images and captions for illustration only; actual size and stimuli as described in the text. Panel B: Number of distinct training stimuli (excluding trials that were one-back targets or during which the subject pressed the response button). There was an additional set of 140 stimuli (70 images and 70 captions) used for testing. Panel C: Setup of the fMRI experiment for the imagery trials. Subjects were instructed to remember 3 image descriptions with corresponding indices (numbers 1 to 3). One of these indices was displayed during the instruction phase, followed by a fixation phase, and then the subjects were imagining the visual scene for 10s.

Images and captions were taken from the training and validation sets of the COCO dataset (Lin et al., 2014). This dataset contains 5 matching captions for each image, of which we only considered the shortest one in order to fit on the screen and to ensure a comparable length for all captions. Spelling errors were corrected manually. As our training set, a random subset of images and another random subset of captions were selected for each subject. All these stimuli were presented only a single time. Information on the number of training stimuli for each subject is shown in **Figure 1** Panel B. Additionally, a shared subset of 140 stimuli (70 images and 70 captions) was presented repeatedly to each subject in order to reduce noise, serving as our test set (on average: 26 times, min: 22, max: 31). Contrary to the training stimuli which were randomly selected from the COCO dataset, the 70 test stimuli were chosen by hand to avoid including multiple scenes that could match the same semantic description. The 70 chosen images as well as their 70 corresponding captions constituted the test set. These stimuli were inserted randomly between the training stimuli.

Note that for each stimulus presented to the subject (e.g. an image), we also have access to the corresponding stimulus in the other modality (the corresponding caption from the COCO dataset), allowing us to estimate model features based on both modalities (vision model features extracted from the image and language model features extracted from the corresponding caption) as well as multimodal features extracted from both the image and the caption.

In addition to these perceptual trials, there were 3 imagery trials for each subject (see also **Figure 1** Panel C). Prior to the first fMRI scanning session, each subject was presented with a set of 20 captions (manually selected to be diverse and easy to visualize) that were not part of the perceptual trials, and they selected 3 captions for which they felt comfortable imagining a corresponding image. Then, they learned a mapping of each caption to a number (1, 2, and 3) so that they could be instructed to perform mental imagery of a specific stimulus, without having to present them with the caption again. The imagery trials occurred every second run, either at the beginning or the end of the run, so that each of the 3 imagery conditions were repeated on average 26 times (min: 23, max: 29). At the start of the imagery trial, the imagery instruction number was presented for 2s, then there was a 1s fixation period followed by the actual imagery period during which a light gray box was depicted for 10s on a dark gray background (the same background that was also used for trials with perceptual input). The light gray box was meant to represent the area in which the mental image should be “projected”. At the end of the experiment, the subjects drew sketches of the images they had been imagining during the imagery trials.

fMRI Preprocessing

Preprocessing of the fMRI data was performed using SPM12 (Ashburner et al., 2014) via nipy (Gorgolewski et al., 2011). We applied slice time correction and realignment for each subject. Each session was coregistered with an anatomical scan of the respective subject’s first session (down-sampled to 2mm³). We created and applied explicit gray matter masks for each subject based on their anatomical scans using a maximally lenient threshold (probability>0).

In order to obtain beta-values for each stimulus, for each subject we fit a GLM (using SPM12) on data from all sessions. We included regressors for train images, train captions, test images, test captions, imagery trials, fixations, blank screens, button presses, and one-back target trials. One-back target trials as well as trials in which the participant pressed the button were excluded in the calculation of all training and test stimulus betas. As output of these GLMs we obtained beta-values for each training and test caption and image as well as the imagery trials.

Finally, we transformed the volume-space data to surface space Freesurfer (Fischl, 2012). We used trilinear interpolation and the fsaverage template in the highest possible resolution (163,842 vertices on each hemisphere) as target.

Modality-Agnostic Decoders

The multimodal nature of our dataset allowed for the training of modality-agnostic decoders. We trained decoders by fitting ridge regression models that take fMRI beta-values as input and predict latent representations extracted from a pretrained deep learning model. Further details on decoder training can be found in [Figure 2](#) as well as Appendix 1.

While modality-specific decoders are trained only on brain imaging data of a single modality, modality-agnostic decoders are trained on brain imaging data from multiple modalities and therefore allow for decoding of stimuli irrespective of their modality.

More specifically, in our case the modality-specific decoders are trained on fMRI beta-values from one stimulus modality, e.g. when subjects were watching images (cf. [Figure 2](#) panel A). Conversely, modality-agnostic decoders are trained jointly using fMRI data from both stimulus modalities (images and captions; cf. [Figure 2](#) panel B). For all decoders, the features that serve as regression targets can either be unimodal (e.g. extracted from images using a vision model) or multimodal (e.g. extracted from both stimulus modalities using a multimodal model, cf. [Figure 2](#) panel C).

We considered features extracted from a range of vision, language, and multimodal models: For vision features, we considered ResNet ([He et al., 2016](#)), ViT ([Dosovitskiy et al., 2020](#)), and DINOv2 ([Oquab et al., 2023](#)); for language features BERT ([Devlin et al., 2019](#)), GPT2 ([Radford et al., 2019](#)), Llama2 ([Touvron et al., 2023](#)), mistral and mixtral ([Jiang et al., 2023](#)). Regarding multimodal features, we extracted features from VisualBERT ([Li et al., 2019](#)), BridgeTower ([Xu et al., 2023](#)), ViLT ([Kim et al., 2021](#)), CLIP ([Radford et al., 2021](#)), ImageBind ([Girdhar et al., 2023](#)), Flava ([Singh et al., 2022](#)), Blip2 ([Li et al., 2023](#)), SigLip ([Zhai et al., 2023](#)), and Paligemma2 ([Steiner et al., 2024](#)). In order to estimate the effect of model training, we further extracted features from a randomly initialized ImageBind model as a baseline. Further details on feature extraction can be found in Appendix 1. All decoders were evaluated on the held-out test data (140 stimuli, 70 captions and 70 images) using pairwise accuracy calculated using cosine distance. Prior to calculating the pairwise accuracy, the model predictions for all stimuli were standardized to have mean of 0 and standard deviation of 1. In the case of imagery decoding, the model predictions were standardized separately. In the case of cross-modal decoding (e.g. mapping an image stimulus into the latent space of a language model), a trial was counted as correct if the caption corresponding to the image (according to the ground-truth in COCO) was closest.

[Figure 3](#) provides an overview on the evaluation metrics. A modality-specific decoder for images can be evaluated on its ability to decode images (Panel A, top) and in a cross-decoding setup for captions (Panel B, bottom). In the same way, we can compute the respective evaluation metrics for modality-specific decoders trained on captions. For the case of modality-agnostic decoders, we evaluate performance for decoding both images and captions using the same single decoder that is trained on both modalities (Panel C).

Results

Modality-Agnostic Decoders

We first compared the performance of modality-specific and modality-agnostic decoders that are trained on the whole brain fMRI data based on different unimodal and multimodal features. The average pairwise accuracy scores are presented in [Figure 4](#). [Figure 5](#) presents pairwise accuracy scores separately for decoding images and for decoding captions. Results for individual subjects can be found in Appendix 5.

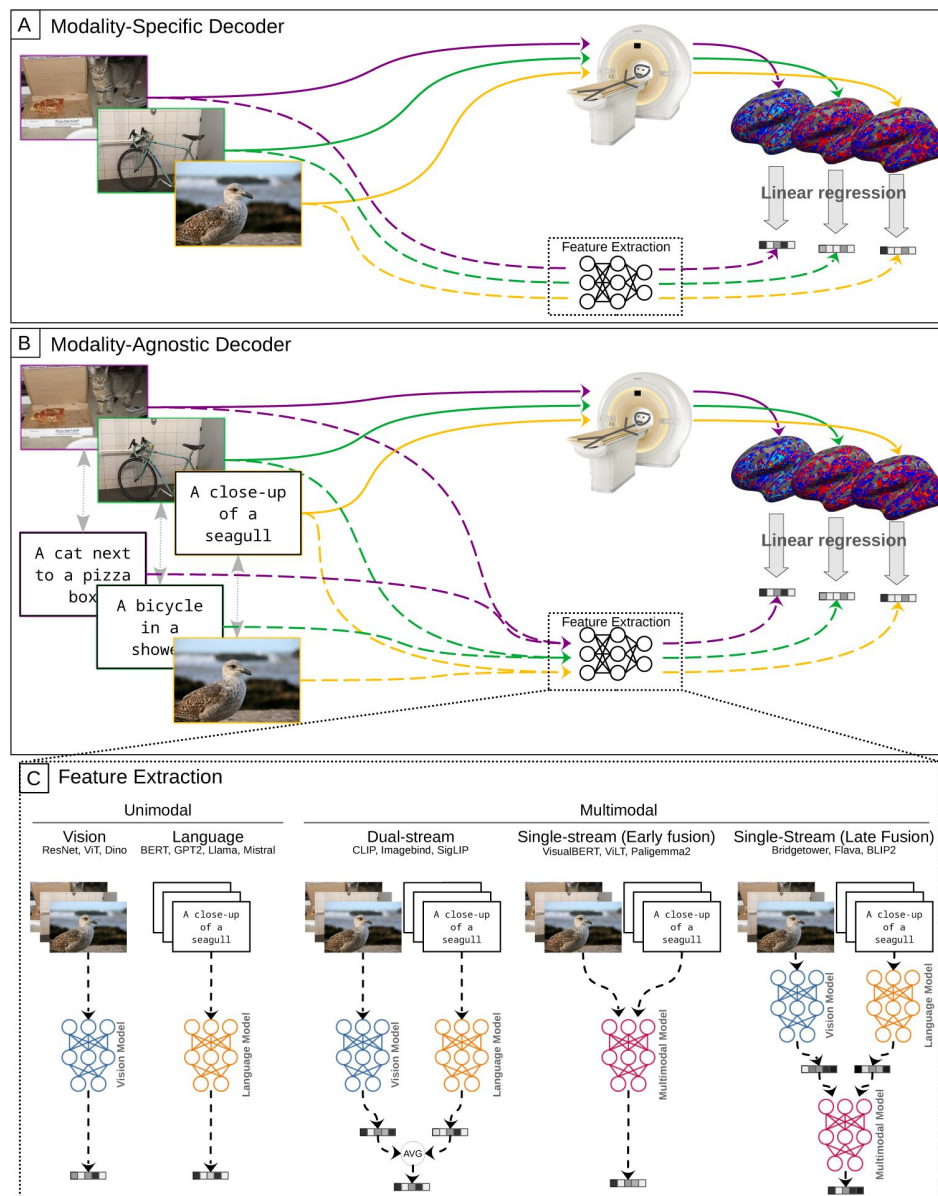


Figure 2.

Training of modality-specific and modality-agnostic decoders.

Panel A: Modality-specific decoders are trained on fMRI data of one modality (e.g. subjects viewing images) by mapping it to features extracted from the same stimuli. Panel B: Modality-agnostic decoders are trained jointly on fMRI data of both modalities (subjects viewing images and captions). Panel C: To train decoders, features can be either extracted unimodally from the corresponding images or captions, or by creating multimodal features based on both modalities. For example, to train a modality-agnostic decoder based on features from a unimodal language model, we map the fMRI data of subjects viewing captions to features extracted from the respective captions using this language model, as well as the fMRI data of subjects viewing images to features extracted by the language model from the corresponding captions. We can also train modality-specific decoders on features from another modality, for example by mapping fMRI data of subjects viewing images to features extracted from the corresponding captions using a language model (cf. crosses on orange bars in [Figure 4](#)) or using multimodal features (cf. crosses on blue bars in [Figure 4](#)).

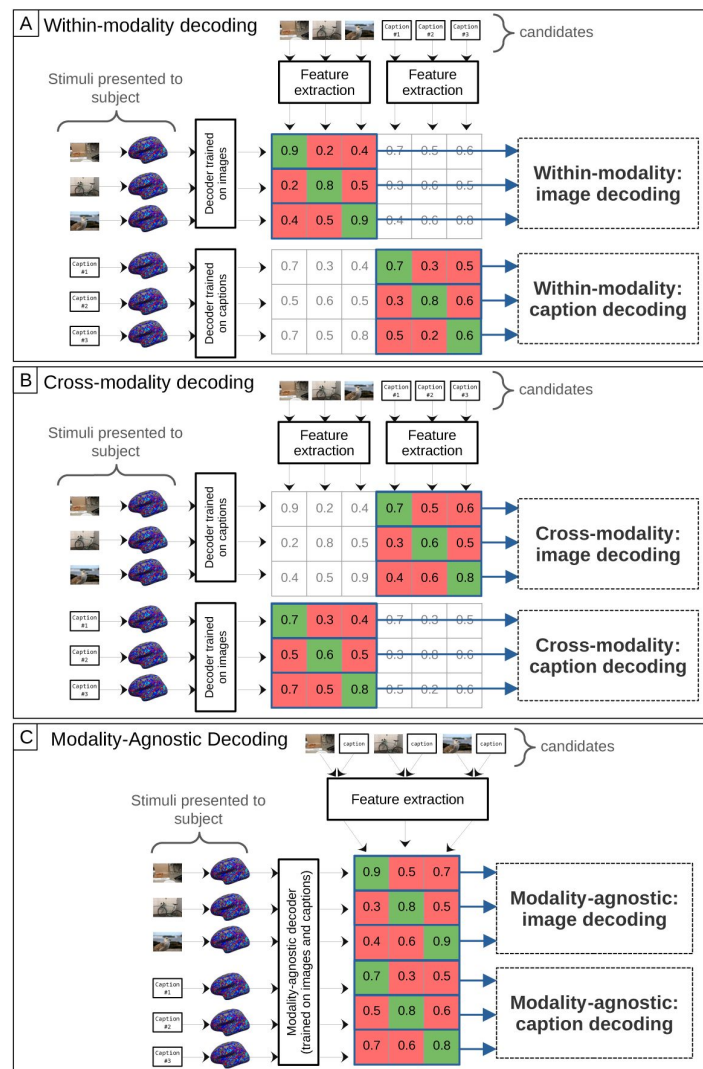


Figure 3.

Evaluation of modality-specific and modality-agnostic decoders.

The matrices display cosine similarity scores between features extracted from the candidate stimuli and features predicted by the decoder. The evaluation metric is pairwise accuracy, which is calculated row-wise: For a given matrix row, we compare the similarity score of the target stimulus on the diagonal (in green) with the similarity scores of all other candidate stimuli (in red). Panel A: Within-modality decoding metrics of modality-specific decoders. To compute within-modality accuracy for image decoding, a modality-specific decoder trained on images is evaluated on all stimuli that were presented as images. To compute within-modality accuracy for caption decoding, a modality-specific decoder trained on captions is evaluated on all caption stimuli. Panel B: Cross-modality decoding metrics of modality-specific decoders. To compute cross-modality accuracy for image decoding, a modality-specific decoder trained on captions is evaluated on all stimuli that were presented as images. To compute cross-modality accuracy for caption decoding, a modality-specific decoder trained on images is evaluated on all caption stimuli. Panel C: Metrics for modality-agnostic decoders. To compute modality-agnostic accuracy for image decoding, a modality-agnostic decoder is evaluated on all stimuli that were presented as images. The same decoder is evaluated on caption stimuli to compute modality-agnostic accuracy for caption decoding. Here we show feature extraction based on unimodal features for modality-specific decoders and based on multimodal features for the modality-agnostic decoder, in practice the feature extraction can be unimodal or multimodal for any decoder type (see also [Figure 2](#)).

Figure 4.

Average decoding scores for modality-agnostic decoders (bars), compared to modality-specific decoders trained on subjects viewing captions (-) or on data from subjects viewing images (x).

The metric is pairwise accuracy (see also [Figure 3](#)). Error bars indicate 95% confidence intervals for modality-agnostic decoders. Chance performance is at 0.5.

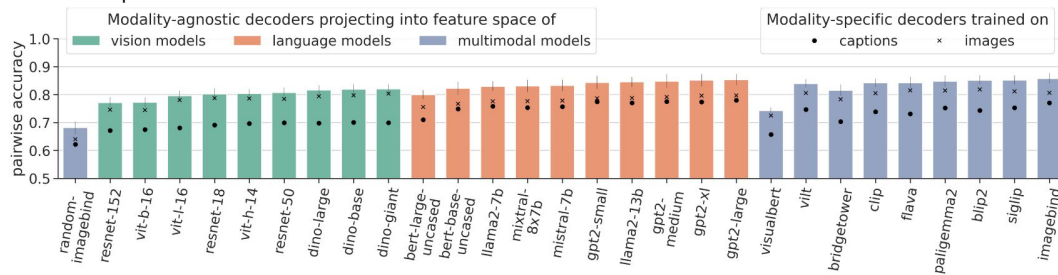
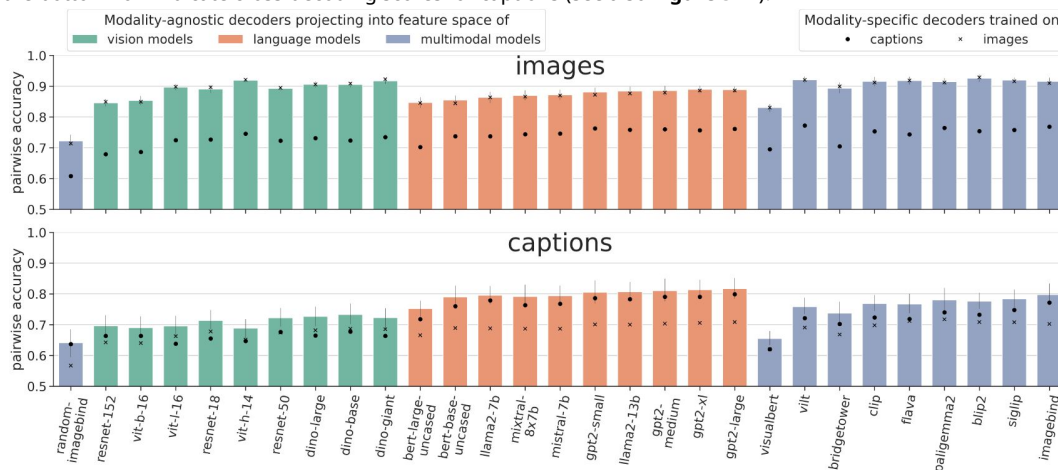


Figure 5.

Decoding accuracy for decoding captions (top) and for decoding images (bottom).

The bars indicate modality-agnostic decoding accuracy. The crosses (x) in the top row and the dots (•) in the bottom row indicate within-modality decoding scores. The dots (•) in the top row indicate cross-decoding scores for images, the crosses (x) in the bottom row indicate cross-decoding scores for captions (see also [Figure 3](#)).



When analyzing the average decoding accuracy (**Figure 4**), we find that modality-agnostic decoders perform better than modality-specific decoders, irrespective of the features that the decoders were trained on. This high performance (which can be attributed to the large training dataset used by modality-agnostic decoders) is achieved despite the additional challenge of not knowing the modality of the stimulus the subject was seeing.

Further, we observed that modality-agnostic decoders based on the best multimodal features (imagebind: $85.71\% \pm 2.58\%$) do not perform substantially better than decoders based on the best language features (GPT2-large: $85.31\% \pm 2.83\%$) and only slightly better than decoders trained on the best vision features (Dino-giant: $82.02\% \pm 2.43\%$). This result suggests that high-performing modality-agnostic decoders do not necessarily need to rely on multimodal features; features extracted from language models can lead to equally high performance. When comparing the different architecture types of models for multimodal feature extraction (dual stream vs. single stream with early fusion vs. single stream with late fusion; cf. Panel C in **Figure 2**), we only observed a slight performance disadvantage for single-stream models with early fusion (Mean accuracy values for dual stream models: 85.04%; for single stream models with early fusion: 81.01%; for single stream models with late fusion 83.63%). We performed a repeated measures ANOVA (grouping the data by subject), comparing the decoding accuracy values of modality-agnostic decoders based on different families of multimodal features. The only significant effect was: `model_family_single_stream_early_fusion` : $\beta = -0.04$, $SE = 0.011$, $p < 1e - 3$.

When analyzing the performance specifically for decoding images (**Figure 5**, top), we find that modality-agnostic decoders perform as well as the modality-specific decoders trained on images (crosses in top row are at the same level as the bars in **Figure 5**; we find no statistically significant difference in their performances. We performed a repeated measures ANOVA (grouping the data by subject), comparing the image decoding accuracy values of modality-agnostic decoders with those of modality-specific decoders trained on images. The resulting p-value for the effect of the `decoder_type` was $p = 0.73$.) Further, modality-agnostic decoders even outperform modality-specific decoders trained on captions for decoding captions (dots in the bottom row are lower than bars in **Figure 5**). We performed a repeated measures ANOVA (grouping the data by subject), comparing the caption decoding accuracy values of modality-agnostic decoders with those of modality-specific decoders trained on captions. The result was: `decoder_type`: $\beta = 0.036$, $SE = 0.006$, $p < 1 \cdot 10^{-8}$. In other words, even if we know that a brain pattern was recorded in response to the subject reading a caption, we are more likely to decode it accurately if we choose to apply a decoder trained using both modalities, than if we apply the appropriate decoder, trained only on captions.

Furthermore, we found that the cross-modal decoding performance for decoding visual stimuli (images) using decoders trained on linguistic stimuli (captions) is higher than the cross-modal decoding performance in the other direction, corroborating similar results from Tang et al., 2023a on movies and audio books (dots in top row are higher than crosses in bottom row in **Figure 5**).

Qualitative Decoding Results

To obtain a better understanding of the decoding performance of the modality-agnostic decoders, we inspected the decoding results for 5 randomly selected test stimuli. We created a large candidate set of 41,118 stimuli by combining the test stimuli and the training stimuli from all subjects. For each stimulus, we ranked these candidate set stimuli based on their similarity to the predicted feature vector. As the test stimuli were shared among all subjects, we could average the prediction feature vectors across subjects to obtain the best decoding results.

Figure 6 presents the results for decoding images using a modality-agnostic decoder trained on ImageBind features. We display the target stimulus along with the top-5 ranked test stimuli. We can observe some clear success cases (the train in the first row) but also failure cases (the teddy

bear decoded as pizza). For the other stimuli, some aspects such as the high-level semantic class (e.g. vehicle, animal) are correctly decoded: For the cars on the highway (last row), the top-ranked images depict trains, which are also vehicles. For the dog (2nd row), the top images contain cats of similar colors. Regarding the giraffe (3rd row), the model appears to have picked up on the fact that there was a body of water depicted in the image.

Note that these qualitative results are not directly comparable with previous work on retrieval or reconstruction using the NSD dataset (Allen et al., 2022 [↗](#); Lin et al., 2022 [↗](#); Takagi and Nishimoto, 2023 [↗](#); Ozelik and VanRullen, 2023 [↗](#)), as our data was collected on a 3T MRI scanner with lower signal-to-noise-ratio than NSD's 7T MRI scanner.

The ranking results for decoding captions are depicted in **Figure 7** [↗](#). The results are somewhat similar to the image decoding results, stimuli that were decoded successfully when presented as image such as the train are also decoded successfully when presented as caption; cases that were failures in the case of image decoding (e.g. the teddy bear) also fail here. However, the top-ranked stimuli for the dog (2nd row) do not always contain animals (the decoder seems to have picked up on the presence of a vehicle in the caption), but the cars on the highway (last row) get decoded rather successfully.

We additionally provide qualitative results for modality-specific decoders in Appendix 2. These results generally reflect the observations from the quantitative results: Modality-agnostic decoders perform similarly to modality-specific decoders evaluated in a within-modality decoding setup, but substantially better than modality-specific decoders when evaluated in cross-decoding setups.

Modality-Agnostic Regions

To provide insight into the spatial organization of modality-agnostic representations in the brain, we performed a surface-based searchlight analysis.

Modality-agnostic regions should contain patterns that generalize between stimulus modalities. Therefore, such regions should allow for decoding of stimuli in both modalities using a decoder that is trained to pick up on modality-agnostic features, i.e. the decoding performance for images and captions of a modality-agnostic decoder should both be above chance. However, as a modality-agnostic decoder is trained on stimuli from both modalities, it could have learned to leverage certain features to project stimuli from one modality and different features to project stimuli from the other modality. We added two conditions to control that the representations directly transfer between the modalities by additionally training two modality-specific decoders and evaluating them according to their cross-decoding performance, i.e. we require that their decoding performance in the modality they were not trained on is above chance. These four conditions are summarized at the top of **Figure 8** [↗](#).

We used ImageBind features for these searchlight analyses as they led to the highest decoding performance when using the whole brain data. The decoders were trained based on the surface projection of the fMRI beta-values. For each vertex, we defined a searchlight with a fixed size by selecting the 750 closest vertices, corresponding to an average radius of ~ 9.4mm. Details on how this size was selected are outlined in Appendix 4.

We trained and evaluated a modality-agnostic decoder and modality-specific decoders for both modalities on the beta-values for each searchlight location and each subject, providing us with a decoding accuracy scores for each location on the cortex. Then we performed t-tests to identify locations in which the decoding performance is above chance ($acc > 0.5$). We aggregated all 4 comparisons by taking the minimum of the 4 t-values at each spatial location. Finally, we performed a threshold-free cluster analysis (TFCE, Smith and Nichols, 2009 [↗](#)) to identify modality-agnostic ROIs (**Figure 8** [↗](#), bottom). We used the default hyperparameters of $h = 2$ and $e = 1$ for surface-based TFCE (Jenkinson et al., 2012 [↗](#)).

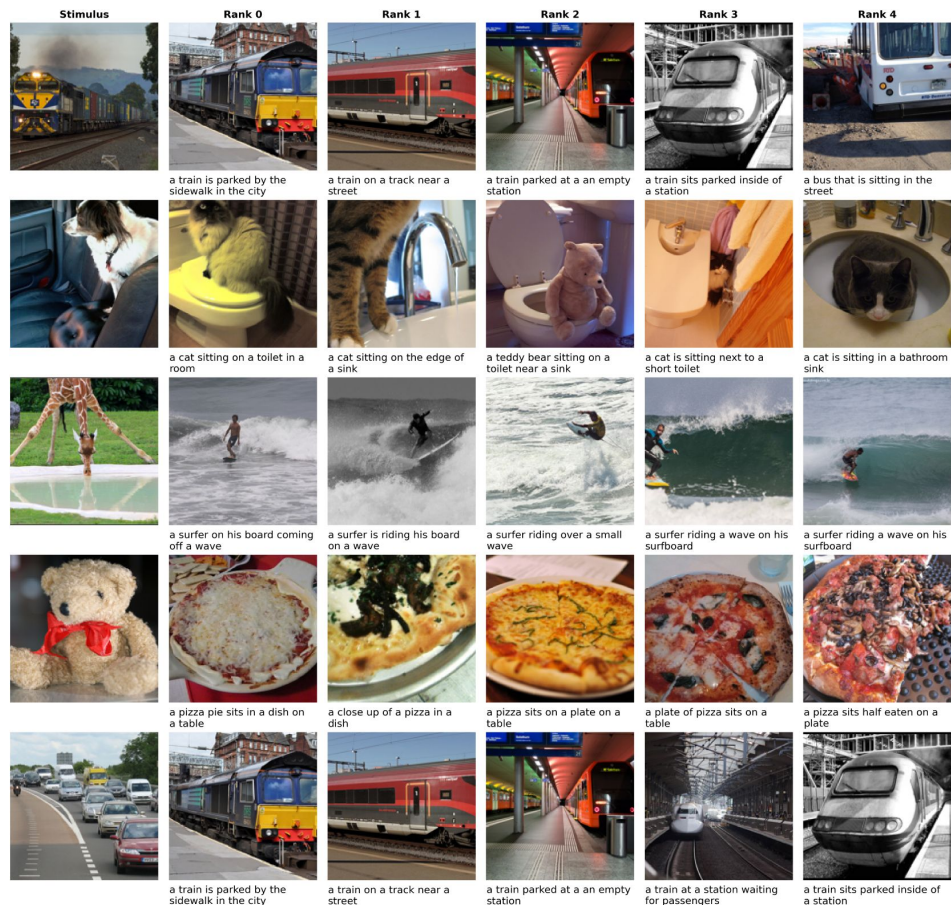


Figure 6.

Decoding examples for image decoding using a modality-agnostic decoder.

The first column shows the image the subject was seeing and the 5 following columns show the candidate stimuli with highest similarity to the predicted features, in descending order. We display both the image and the caption of the candidate stimuli because the decoder is based on multimodal features that are extracted from both modalities. All images were taken from the CoCo dataset (Lin et al., 2014 [\[14\]](#)).

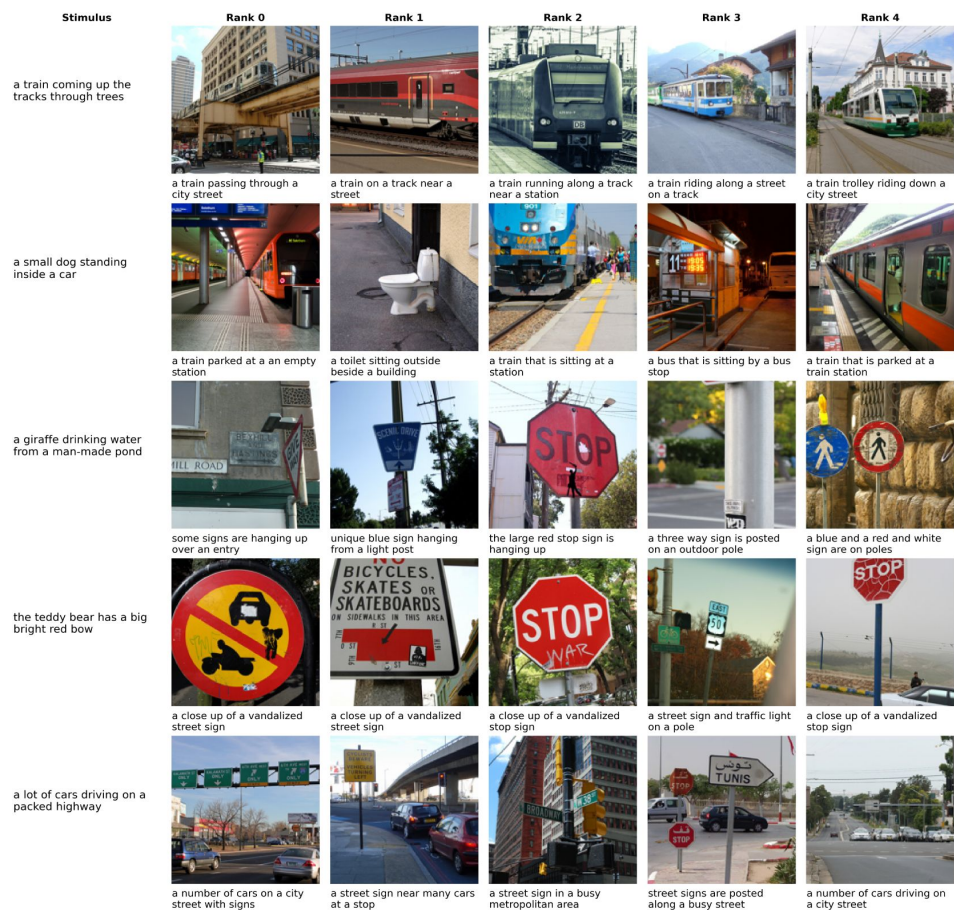


Figure 7.

Decoding examples for caption decoding using a modality-agnostic decoder.

For details see caption of [Figure 6](#). All images were taken from the CoCo dataset (Lin et al., 2014).

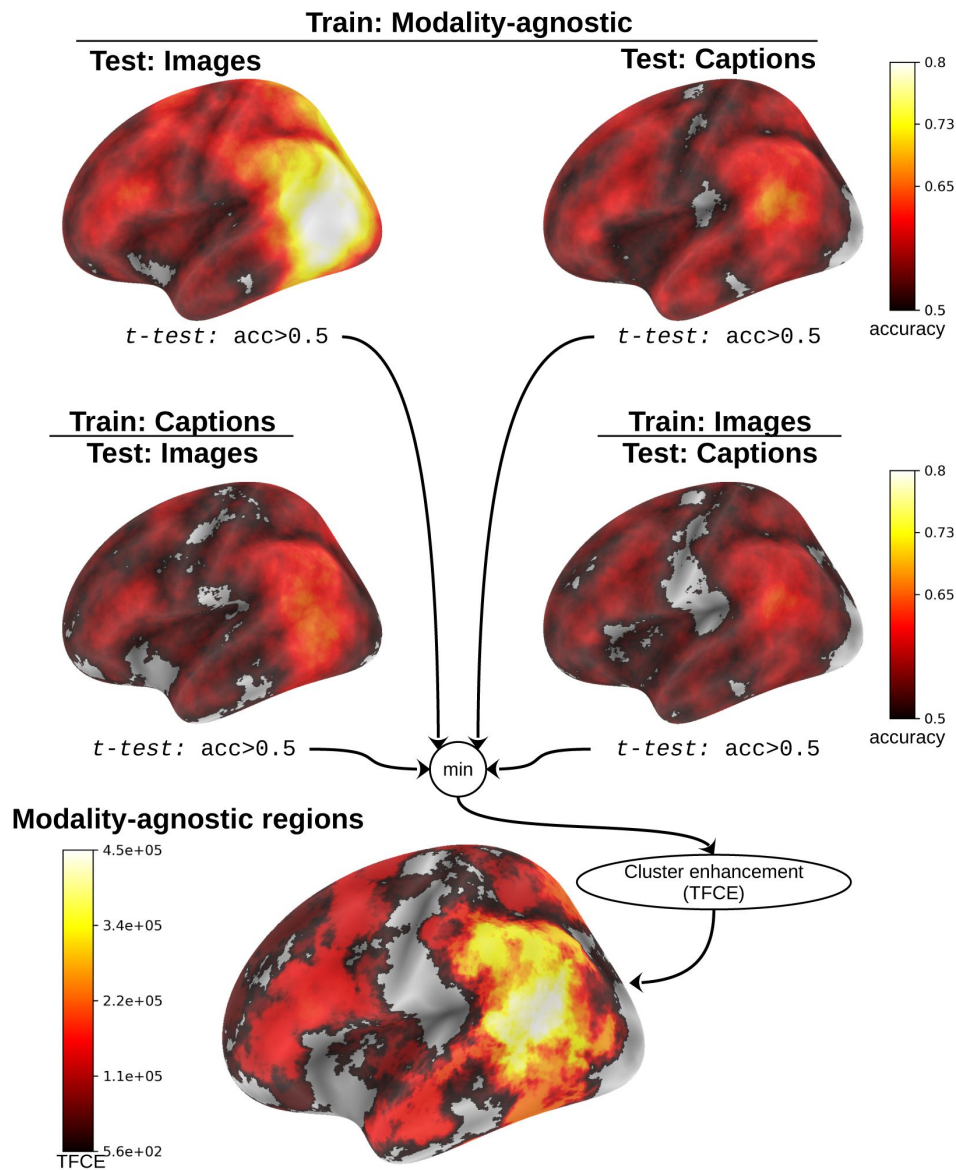


Figure 8.

Searchlight method to identify modality-agnostic ROIs.

The top plots show performance (pairwise accuracy averaged over subjects) of modality-agnostic decoders for decoding images (top left) and decoding captions (top right). In the second row, we display cross-decoding performances: On the left, modality-specific decoders trained on captions are evaluated on images. On the right, modality-specific decoders trained on images are evaluated on captions. We identified modality-agnostic ROIs as clusters in which all 4 decoding accuracies are above chance by taking the minimum of the respective t -values at each location, then performed TFCE to calculate cluster values. The plot only shows left medial views of the brain to illustrate the method, different views of all resulting clusters are shown in [Figure 9](#).

To estimate the statistical significance of the resulting clusters we performed a permutation test. For each subject, we evaluated the decoders 100 times with shuffled labels to create a surrogate distribution. Then, we sampled 10,000 permutations of the 6 subjects' surrogate distributions and calculated group-level statistics (TFCE values) for each of them. Based on this null distribution, we calculated p-values for each cluster. To control for multiple comparisons across space, we took the maximum TFCE score across vertices for each permutation (Smith and Nichols, 2009 [↗](#)).

The results of the surface-based searchlight analysis are presented in **Figure 9** [↗](#). The analysis revealed that modality-agnostic patterns are actually widespread across the brain, especially on the left hemisphere. Peak cluster values were found in the left supramarginal gyrus, inferior parietal gyrus and posterior superior temporal sulcus. Regions belonging to the precuneus, isthmus of the cingulate gyrus, parahippocampus, middle temporal gyrus, inferior temporal gyrus and fusiform gyrus also showed high cluster values.

Imagery Decoding

Finally, we evaluated the ability of decoders trained on the fMRI data with perceptual input to decode stimuli during the imagery conditions.

For a modality-agnostic decoder trained on the whole-brain data, the imagery pairwise decoding accuracy reaches 84.48% (averaged across subjects and model features) when using the 3 imagery stimuli as the candidate set. Note that we used the ground-truth caption and corresponding image from COCO in this candidate set, and not the sketches drawn by the subjects. When the whole test set is added to the candidate set (in total: 73 stimuli), the average pairwise accuracy drops to 72.47%. This substantial drop in performance is most likely explained by the fact that the predicted features for the imagery trials were standardized using only 3 stimuli, and this transformation emphasized differences that enabled distinguishing the 3 imagery trials but do not generalize to the larger test set. We also attempted decoding without standardization of the predicted feature vectors, but this led to much lower performance.

As expected, we also found that modality-agnostic decoders are better suited for imagery decoding than modality-specific decoders. We compared the imagery decoding accuracy of both decoder types taking into account the results for all features and all subjects. To this end, we performed two repeated measures ANOVAs (grouping the data by subject), once comparing the accuracy values of modality-agnostic decoders with those of modality-specific decoders trained on images, and once comparing modality-agnostic decoders to modality-specific decoders trained on captions. The average decoding accuracies were 69.42% for a modality-specific decoder trained on images and 70.02% for a modality-specific decoder trained on captions (vs. 72.47% for a modality-agnostic decoder, as mentioned above). In both comparisons, the accuracy values for the two decoder types were significantly different (when comparing modality-agnostic decoders to modality-specific decoders trained on images: $\text{decoder_type} : \beta = 0.03, SE = 0.011, p < 0.01$; and when comparing to modality-specific decoders trained on captions: $\text{decoder_type} : \beta = 0.024, SE = 0.012, p < 0.04$).

Appendix 6 presents qualitative decoding results for the imagery trials for each subject as well as the sketches of the mental images drawn at the end of the experiment. As expected, the results are worse than those for perceived stimuli, but for several subjects it was possible to decode some major semantic concepts.

We further computed the imagery decoding accuracy during the searchlight analysis. **Figure 10** [↗](#) shows the result clusters for decoding imagery (using the whole test set + the 3 imagery trials as potential candidates).

We observe that many regions that were found to contain modality-agnostic patterns (cf. **Figure 9** [↗](#)) are also regions in which decoding of mental imagery is possible.

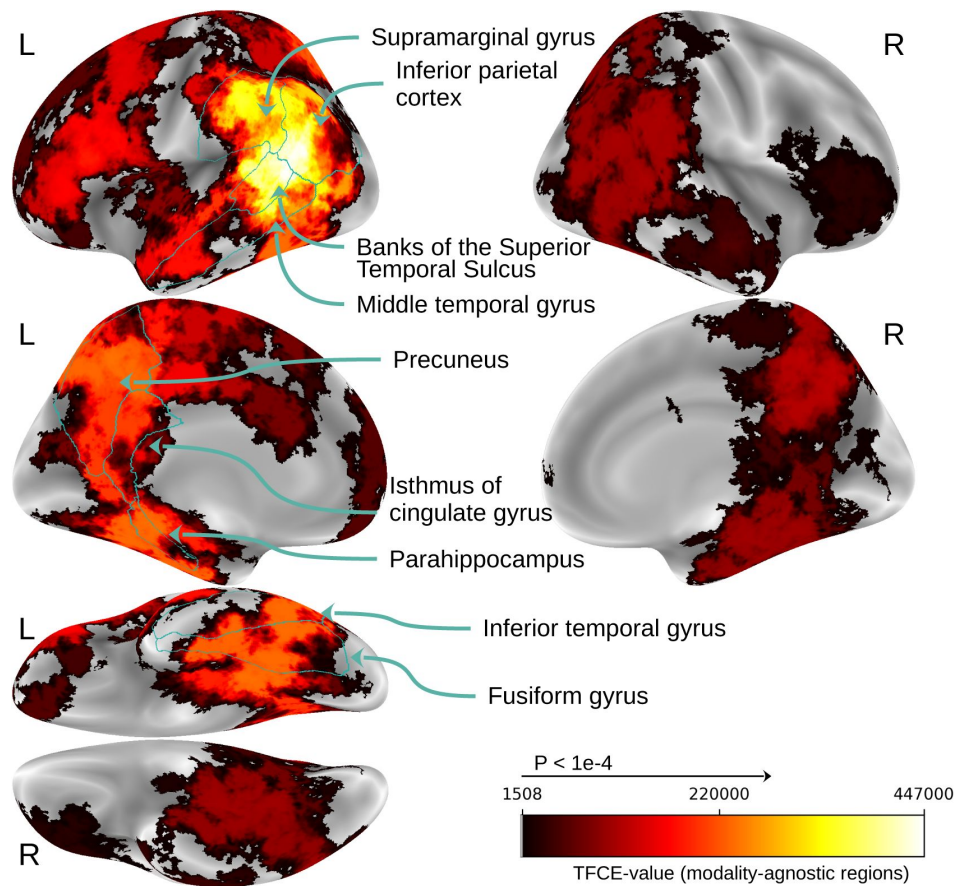


Figure 9.

Searchlight results for modality-agnostic regions.

Maps thresholded at TFCE value of 1508, which is the significance threshold value for which $p < 10^{-4}$ based on the permutation testing. Regions with highest cluster values are outlined and annotated based on the Desikan-Killiany atlas (Desikan et al., 2006 [\[1\]](#)).

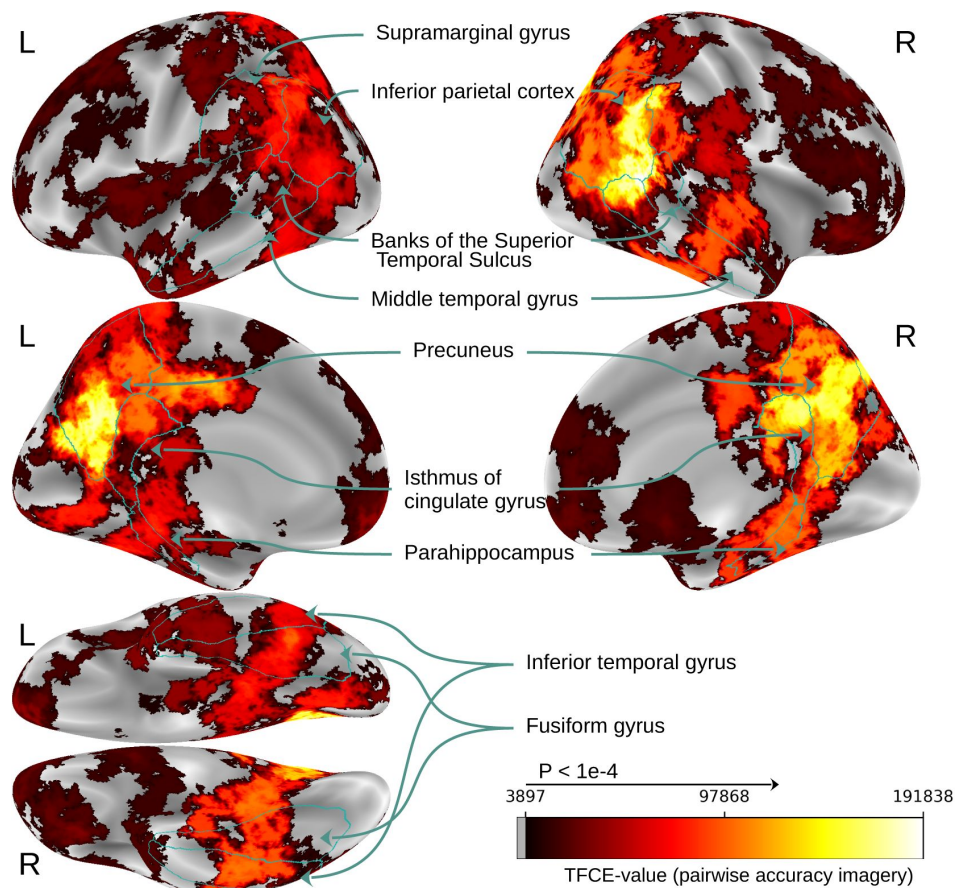


Figure 10.

Searchlight results for imagery decoding.

Maps thresholded at TFCE values that surpass the significance threshold of $p < 10^{-4}$ based on a permutation test. Maps thresholded at TFCE value of 3897, which is the significance threshold value for which $p < 10^{-4}$ based on the permutation testing. We used the pairwise accuracy for imagery decoding using the large candidate set of 73 stimuli. We outlined the same regions as in [Figure 9](#) to facilitate comparison.

One main difference is that the imagery decoding clusters appear to be less left-lateralized than the modality-agnostic region clusters (peak cluster values can be found both on the right inferior parietal cortex and bilaterally in the precuneus). To estimate overlap of the regions allowing for imagery decoding and modality-agnostic regions we calculated the correlation between the TFCE values that were used for identifying modality-agnostic regions (**Figure 9**) and the TFCE values for imagery decoding (**Figure 10**). The Pearson correlation score for the left hemisphere is 0.41 ($p < 1e - 8$), and for the right hemisphere 0.62 ($p < 1e - 8$). Importantly, these correlation scores are substantially higher when compared to the correlation with decoding accuracy of modality-specific decoders: The correlation between the TFCE values for imagery decoding and TFCE values for image decoding of a modality-specific decoder trained on images is 0.28 on the left hemisphere and 0.40 on the right hemisphere. When using TFCE values based on the caption decoding accuracy of a modality-specific decoder trained on captions we obtain 0.19 on the left hemisphere and 0.45 on the right hemisphere.

Discussion

In this work, we introduced a new large-scale multimodal fMRI dataset that enables the development of models for modality-agnostic decoding of visual and linguistic stimuli using a single model. These modality-agnostic decoders were specifically trained to pick up on modality-agnostic patterns, enabling a performance increase over modality-specific decoders when decoding linguistic stimuli in the form of captions.

According to a range of theories, modality-agnostic representations are tightly linked to (lexical-) semantic representations (Simmons and Barsalou, 2003; Binder et al., 2009; Meschke and Gallant, 2024). Most importantly, a range of studies that aimed to identify brain regions linked to semantic/conceptual representations by asking subjects to perform tasks that require semantic processing of words found evidence for such regions that overlap to a high degree with the regions identified in our study (Fernandino et al., 2016; Martin et al., 2018; Carota et al., 2021; Fernandino et al., 2022; Tong et al., 2022). A strong link between these systems could also explain our result that modality-agnostic decoders based on unimodal representations from language models are performing as well as decoders based on multimodal representations (cf. **Figure 4**), as well as the partial left-lateralization of the identified modality-agnostic regions.

In a second analysis, we additionally leveraged our dataset to localize modality-agnostic regions in the brain by searching for areas in which decoding of both stimulus modalities is possible using modality-agnostic decoders as well as in a cross-decoding setup. This approach lead to the identification of a large network involving temporal, parietal, and frontal regions, and peak cluster values on the left hemisphere (cf. **Figure 9**). All areas with high cluster values confirm findings from previous studies: The left precuneus (Shinkareva et al., 2011; Popham et al., 2021; Handjaras et al., 2016), posterior cingulate/ retrosplenial cortex (Fairhall and Caramazza, 2013; Handjaras et al., 2016), supramarginal gyrus (Shinkareva et al., 2011), inferior parietal cortex (Man et al., 2012; Vandenberghe et al., 1996; Shinkareva et al., 2011; Devereux et al., 2013; Popham et al., 2021; Simanova et al., 2014; Handjaras et al., 2016), superior temporal sulcus (Man et al., 2012), middle temporal gyrus (Vandenberghe et al., 1996; Shinkareva et al., 2011; Fairhall and Caramazza, 2013; Devereux et al., 2013; Handjaras et al., 2016), inferior temporal gyrus (Vandenberghe et al., 1996; Shinkareva et al., 2011; Fairhall and Caramazza, 2013; Simanova et al., 2014; Handjaras et al., 2016), fusiform gyrus (Vandenberghe et al., 1996; Moore and Price, 1999; Bright et al., 2004; Shinkareva et al., 2011; Fairhall and Caramazza, 2013; Simanova et al., 2014), and parahippocampus (Vandenberghe et al., 1996). However, previous studies have led to contradicting results regarding the locality of modality-agnostic regions (they were identifying varying subsets of these regions; see also Appendix 3), probably due to the limited number and artificial nature of stimuli employed. Our method identified *almost all* of the previously proposed

regions as regions with modality-agnostic patterns, highlighting the advantage of this large multimodal dataset in which subjects are viewing photographs of complex natural scenes and reading full English sentences. The left superior occipital gyrus was not identified in our study, but in previous studies by [Vandenberghe et al., 1996](#); [Shinkareva et al., 2011](#). However, we found that a major part of the left superior occipital *sulcus* represents modality-agnostic information. Further, [Jung et al., 2018](#) found modality-agnostic patterns in the right superior frontal gyrus. One major difference between their study and ours is that they used auditory input as a second modality instead of text. Further work is required to investigate to what extent the modality-agnostic regions identified in our work generalize to all modalities.

The fact that the presence of modality-agnostic patterns is positively correlated with the imagery decoding performance in different locations provides further evidence that the identified patterns are truly modality-agnostic. We further found that decoders trained exclusively on data for which participants were exposed to perceptual input do generalize to imagery trials, confirming previous findings that were based on more limited stimulus sets ([Stokes et al., 2009](#); [Reddy et al., 2010](#); [Lee et al., 2012](#); [Johnson and Johnson, 2014](#); [Naselaris et al., 2015](#)). Regarding the representations involved in mental imagery, we found that modality-agnostic decoders outperform modality-specific decoders in terms of imagery decoding. This finding can be seen as support for the involvement of modality-agnostic representations in mental imagery, as modality-agnostic decoders were trained explicitly to pick up on such patterns.

The findings of our searchlight analysis for imagery decoding suggest that mental imagery indeed involves a large network of regions across both hemispheres of the cerebral cortex. This includes high-level visual areas, parietal areas such as the precuneus and inferior parietal cortex and several frontal regions, but also parts of the early visual cortex. Results are highly similar on both hemispheres, highlighting the involvement of large-scale bilateral brain networks during mental imagery of complex scenes.

While there are lesion studies on hemispheric asymmetries that suggest that regions in the left hemisphere are crucial for mental imagery ([Farah, 1984](#); [Bartolomeo, 2002](#)), a more recent review that additionally considers evidence from neuroimaging and direct cortical stimulation studies suggests that frontoparietal networks in both hemispheres are involved in mental imagery, and that lateralization patterns can be found in the temporal lobes ([Liu et al., 2022](#)). Such lateralization was found to depend on the nature of the imagined items, the imagination of objects and words involving the left inferior temporal cortex while the imagination of faces and people was found to be more right-lateralized and the imagination of complex scenes (as in our study) leads to significant activity in both hemispheres ([O’Craven and Kanwisher, 2000](#); [Steel et al., 2021](#); [Spagna et al., 2021](#)). Crucially, in more recent decoding studies, results were either observed bilaterally, or the analyses did not target hemispheric asymmetries ([Reddy et al., 2010](#); [Lee et al., 2012](#)). But see [Stokes et al., 2009](#) in which perception-to-imagery generalization of single letters was left-lateralized. Our work shows for the first time results of a searchlight analysis of imagery decoding of complex visual scenes. Above-chance decoding is possible on both hemispheres, with the highest decoding accuracies in the precuneus and the right inferior parietal cortex and the superior temporal sulcus. Future investigations with larger sets of imagined scenes could address the question whether lateralization patterns depend on the nature of the imagined objects.

It remains an open question whether the activation patterns in the modality-agnostic regions identified in our study relate to abstract concepts or to lower-level features that are shared between the two modalities. [Binder, 2016](#) puts this dichotomy into question, considering that “there is no absolute demarcation between embodied/perceptual and abstract/conceptual representation in the brain.” (p. 1098). The author argues for a hierarchical system in which

representational patterns become increasingly abstract, creating a continuum between actual experiential information up to higher-level conceptual information (see also [Andrews et al., 2014](#) [↗](#)).

According to the results of our searchlight analysis, the anterior temporal lobes are not among the regions with the highest probability of being modality-agnostic, contradicting the hypothesis of the hub-and-spoke theory that these areas are the major semantic hub in the brain. However, MRI signals from these regions have a lower signal-to-noise ratio with standard fMRI pulse sequences ([Devlin et al., 2000](#) [↗](#); [Embleton et al., 2010](#) [↗](#)). A more targeted study with an adapted fMRI protocol would be required to shed light on the nature of patterns in these regions. More generally, the hub-and-spoke theory also puts emphasis on the role of spokes for the formation of conceptual representations ([Pobric et al., 2010](#) [↗](#); [Ralph et al., 2017](#) [↗](#)). Future work could be aimed at testing the hub-and-spoke theory proposal that features in hierarchically lower level representation spaces of the spokes are not directly relatable to features in the representation space of the hubs: Object representations in the spokes are based on interpretable features (e.g. shape, color, affordances of an object) and get translated into another representational format that is representing conceptual similarities (but its dimensions do not directly map to interpretable features) in the semantic hub ([Frisby et al., 2023](#) [↗](#)). To test this hypothesis, modality-agnostic representations in the anterior temporal lobes (measured with targeted fMRI pulse sequences) could be compared to representations in candidate regions for modality-specific spokes using RSA.

The modality-agnostic regions we found in the searchlight analysis can also be seen as candidates for convergence zones, in which increasingly abstract representations are formed ([Damasio, 1989](#) [↗](#); [Tranel et al., 1997](#) [↗](#); [Meyer and Damasio, 2009](#) [↗](#)). To obtain further insight into the hierarchical organization of these zones, future work could take advantage of the improved temporal resolution of other brain imaging techniques such as MEG to explore in which areas modality-agnostic patterns are formed first, and how they are being transformed when spreading to higher-level areas of the brain ([Dirani and Pykkänen, 2024](#) [↗](#); [Benchetrit et al., 2024](#) [↗](#)).

In line with the GRAPES framework, ([Martin, 2009](#) [↗](#), [2016](#) [↗](#)), we found that modality-agnostic representations are distributed across temporal and parietal areas. To test the related hypothesis that conceptual information is organized in domains, we plan to use RSA to understand which kind of semantic information is represented in the different modality-agnostic regions identified.

Finally, our results can be interpreted with respect to the Global Workspace Theory. All modality-agnostic regions are good candidate regions for a global workspace. They could, however, also be part of modality-specific modules that get activated in a modality-agnostic fashion through a “broadcast” operation as a stimulus is perceived consciously ([Baars, 1993](#) [↗](#), [2005](#) [↗](#)). To distinguish these two cases, an experimental manipulation of attention could be used: according to Global Workspace Theory, attention is required for information to enter the workspace, but not for the workspace signals to reach other brain regions via broadcast. In the future, we plan to investigate how modality-agnostic patterns are modulated by attention, by analyzing additional test sessions from the same subjects in which they were instructed in specific runs to pay attention to only one of the modalities. These sessions will be released as part of another future dataset and publication.

To conclude, the results from our searchlight analysis so far are in line with all major theories on modality-agnostic representations that were considered. As this dataset will be shared publicly, more targeted investigations can be performed by the research community in order to adjudicate between different theories.

Acknowledgements

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Appendix 1 Feature extraction details

For each target stimulus (image or caption), our database also contained an equivalent stimulus in the other modality (caption or image). In this way, we could extract model features from the corresponding image for vision models, the corresponding caption for language models, and an multimodal representation of both image and caption for the multimodal models. We used publicly available pretrained models implemented in the HuggingFace Transformers library (Wolf et al., 2020 [🔗](#)) or from their respective authors' repositories.

Model versions for unimodal models are as indicated in **Figure 4** [🔗](#). For multimodal models, the exact version for CLIP was clip-vit-large-patch14, for ViLT vilt-b32-mlm, for Visual-BERT visualbert-nlvr2-coco-pre, for Imagebind imagebind_huge, for Bridgetower bridgetower-large-itm-mlm-itc, for Flava flava-full, for SigLip siglip-so400m-patch14-384, and for Paligemma2 paligemma2-3b-pt-224.

We extracted language features from all models by averaging the outputs for each token, as this was established as common practice for the extraction of sentence embeddings from Transformer-based language models (e.g. Krasnowska-Kieraś and Wróblewska, 2019 [🔗](#); Reimers and Gurevych, 2019 [🔗](#)).

For Transformer-based vision models, we compared representations extracted by averaging the outputs for each patch with representations extracted from [CLS] tokens in Table 1. We found that for almost all models, the mean features allow for higher decoding accuracies. For all experiments reported in the main paper we therefore only considered this method.

For multimodal models, we also compared a range of techniques for feature extraction. The results are shown in Table 2.

For dual-stream multimodal models, we averaged the vision and language features to create the final multimodal feature representation. For single-stream multimodal features we compared using representations extracted by averaging the outputs for each token with representations extracted from [CLS] token and found that the averaged output leads to better performance in almost all cases (see Table 2). Further, some single-stream models (Flava, Paligemma2 and BLIP2) allow for feature extraction based on intermediate vision and language representations in addition to a direct extraction of multimodal features (based on fused representations from the multimodal stream). We found that averaging features from these intermediate vision and language representations leads to better performance (see Table 2). For all results reported in the main paper we used the feature extraction method leading to best performance for each model.

Appendix 1—table 1.

Feature comparison for vision models.

Pairwise accuracy for modality-agnostic decoders based on vision features extracted by averaging the last hidden states of all patches (“vision_features_mean”) compared to when using features extracted from [CLS] tokens (“vision_features_cls”). The method leading to the best decoding performance for each model is highlighted in bold.

model	vision features	pairwise accuracy
dino-base	vision_features_cls	0.763
dino-base	vision_features_mean	0.819
dino-giant	vision_features_cls	0.750
dino-giant	vision_features_mean	0.820
dino-large	vision_features_cls	0.754
dino-large	vision_features_mean	0.816
vit-b-16	vision_features_cls	0.788
vit-b-16	vision_features_mean	0.772
vit-h-14	vision_features_cls	0.785
vit-h-14	vision_features_mean	0.804
vit-l-16	vision_features_cls	0.788
vit-l-16	vision_features_mean	0.796

Appendix 1—table 2.

Feature comparison for multimodal models.

The features are either based on the [CLS] tokens from the fused representations (“fused_cls”), averaging over the fused tokens (“fused_mean”), or averaging over tokens from intermediate vision and language stream outputs (“avg”). For the last case, for some models the vision and language features can also be either based on [CLS] or based on averaging over all tokens/patches. The method leading to the best decoding performance for each model is highlighted in bold.

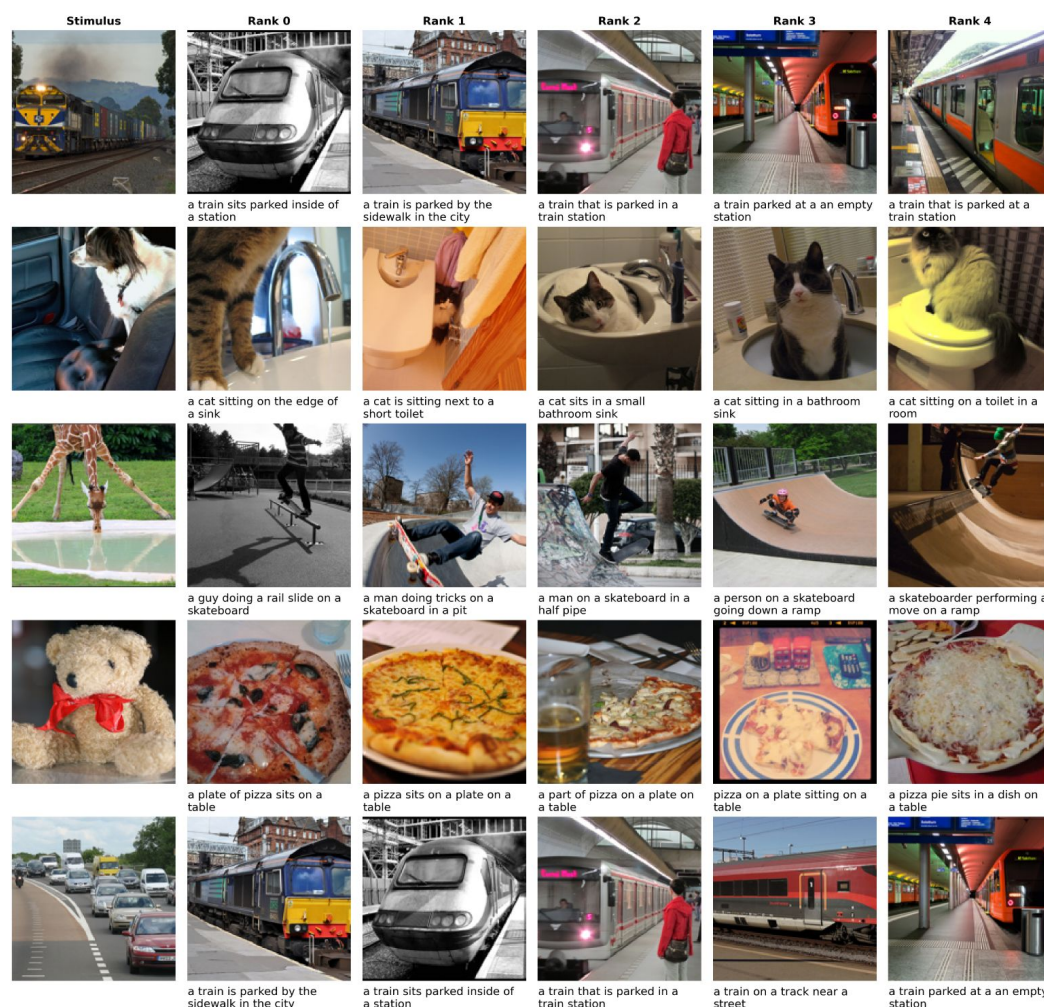
model	features	vision features	language features	pairwise accuracy
blip2	avg	vision_features_cls	lang_features_cls	0.851
blip2	fused_cls	vision_features_cls	lang_features_cls	0.710
blip2	fused_mean	vision_features_cls	lang_features_cls	0.740
bridgetower	fused_cls			0.815
bridgetower	fused_mean			0.788
clip	avg	vision_features_cls	lang_features_cls	0.842
flava	avg	vision_features_cls	lang_features_cls	0.842
flava	fused_cls	vision_features_cls	lang_features_cls	0.752
flava	fused_mean	vision_features_cls	lang_features_cls	0.772
imagebind	avg	vision_features_cls	lang_features_cls	0.857
paligemma2	avg	vision_features_cls	lang_features_mean	0.829
paligemma2	avg	vision_features_mean	lang_features_mean	0.848
paligemma2	fused_mean	vision_features_mean	lang_features_mean	0.828
siglip	avg	vision_features_cls	lang_features_cls	0.852
siglip	avg	vision_features_mean	lang_features_cls	0.823
vilt	fused_cls			0.759
vilt	fused_mean			0.839
visualbert	fused_cls			0.639
visualbert	fused_mean			0.743

Decoder training details

The decoders were linear ridge-regression models as implemented in the scikit-learn library (Pedregosa et al., 2011). All training data was standardized to have mean of 0 and standard deviation of 1. The test data was standardized using the mean and standard deviation of the training data. The regularization hyperparameter α was optimized using 5-fold cross validation on the training set (values considered: $\alpha \in \{1e3, 1e4, 1e5, 1e6, 1e7\}$). Afterwards, a final model was trained using the best α on the whole training set.

Appendix 2 Qualitative Decoding Results for Modality-Specific Decoders

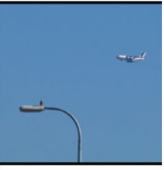
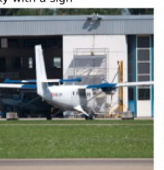
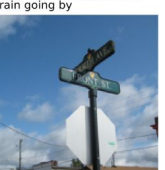
In this section we present qualitative decoding results for modality-specific decoders for the same examples as presented in Section Qualitative Decoding Results for modality-agnostic decoders. The results show that modality-agnostic decoders are as good as modality-specific decoders when evaluated in a within-modality decoding setup (Figures 1 and 3) but substantially better than modality-specific decoders when evaluated in a cross-decoding setup (Figures 2 and 4).



Appendix 2—figure 1.

Decoding examples for image decoding using a modality-specific decoder trained on images (within-modality decoding).

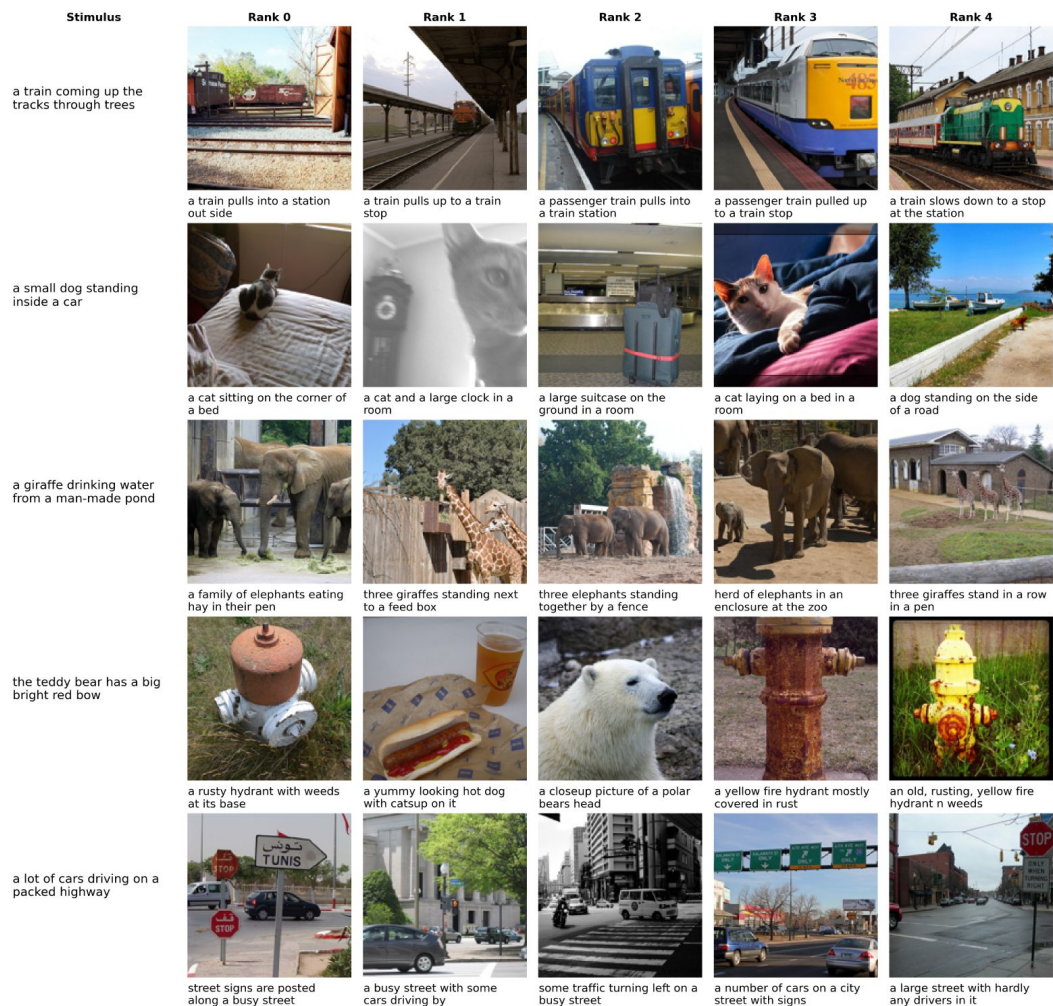
The first column shows the image the subject was seeing and the 5 following columns show the candidate stimuli with highest similarity to the predicted features, in descending order. We display both the image and the caption of the candidate stimuli because the decoder is based on multimodal features that are extracted from both modalities. All images were taken from the CoCo dataset (Lin et al., 2014).

Stimulus	Rank 0	Rank 1	Rank 2	Rank 3	Rank 4
a train coming up the tracks through trees	 an air plane flies low above a highway	 a plane is flying past a street light	 an airplane taking off into the open sky	 a plane is flying through the sky with a sign	 a plane is taking off from an airport runway
a small dog standing inside a car	 an air plane flies low above a highway	 a plane is taking off from an airport runway	 there is a stop sign near a train going by	 an airplane on the ground on a run way	 an air plane sitting on the run way
a giraffe drinking water from a man-made pond	 a plane is flying through the sky with a sign	 a stop sign with a sky in the background	 a street sign on a pole against a blue sky	 a red stop sign against a blue sky	 a stop sign stands out against a blue sky
the teddy bear has a big bright red bow	 a street sign on a pole against a blue sky	 a stop sign with a sky in the background	 a street sign on a city side walk	 a street sign on a city side walk	 a street sign near a traffic light
a lot of cars driving on a packed highway	 an airplane is in flight in the blue sky	 an airplane is flying in the blue sky	 a plane seems to be flying in the blue sky	 an airplane flying through the open blue sky	 an airplane flying high in the blue skies

Appendix 2—figure 2.

Decoding examples for caption decoding using a modality-specific decoder trained on images (cross-modality decoding).

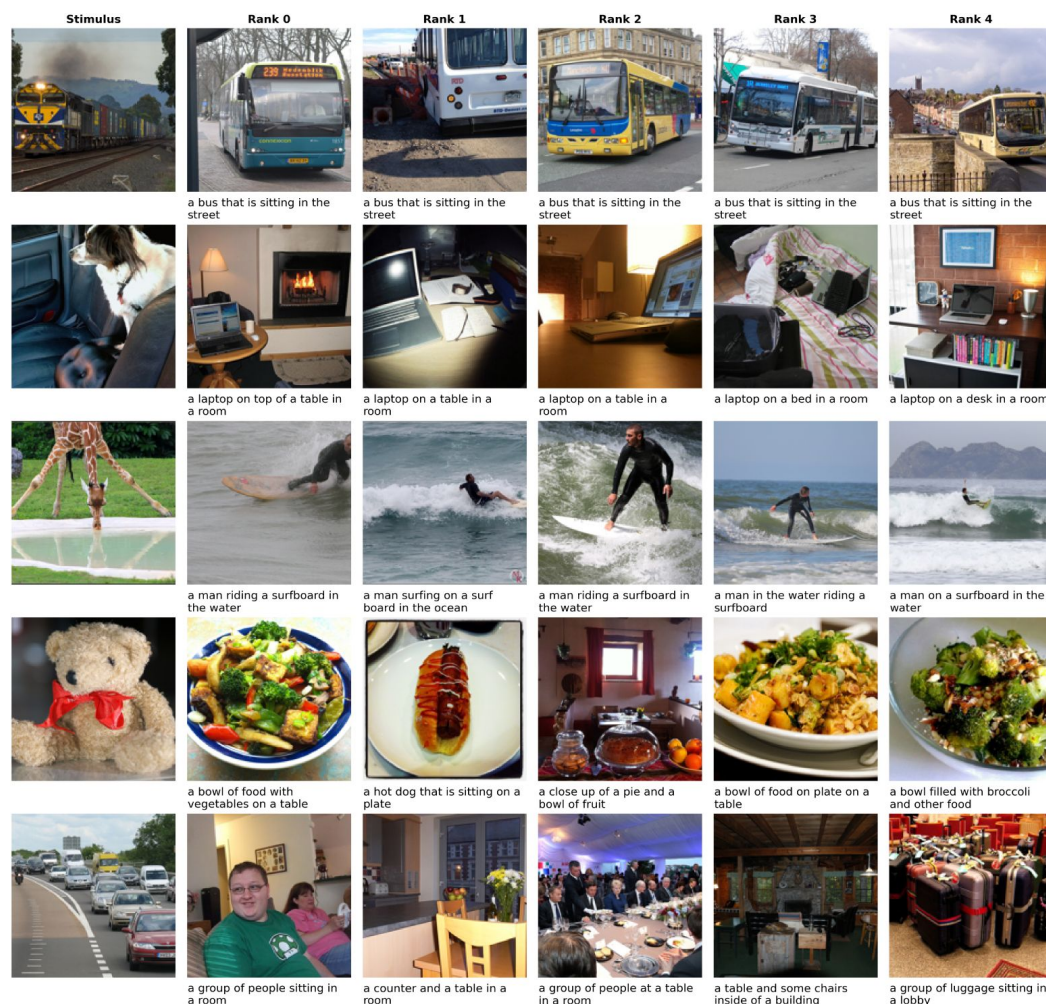
For details see caption of Figure 1. All images were taken from the CoCo dataset (Lin et al., 2014).



Appendix 2—figure 3.

Decoding examples for caption decoding using a modality-specific decoder trained on captions (within-modality decoding).

For details see caption of [Figure 1](#). All images were taken from the CoCo dataset (Lin et al., 2014).



Appendix 2—figure 4.

Decoding examples for image decoding using a modality-specific decoder trained on captions (cross-modality decoding).

For details see caption of [Figure 1](#). All images were taken from the CoCo dataset ([Lin et al., 2014](#)).

Appendix 3 Candidates for Modality-agnostic regions

Table 1 shows candidates for modality-agnostic regions that were identified by previous work. We only considered fMRI experiments involving multiple stimulus modalities. Many other studies relied on unimodal stimuli and semantic tasks to identify modality-agnostic/ conceptual regions, these are however less directly comparable to our setup.

Hemi Region	Studies that identified the region as modality-agnostic
left hemisphere	superior occipital gyrus*
	(Vandenberghe et al., 1996; Shinkareva et al., 2011)
	(Shinkareva et al., 2011)
	middle occipital gyrus
	(Shinkareva et al., 2011; Simanova et al., 2014)
	superior temporal gyrus
	(Shinkareva et al., 2011)
	superior temporal sulcus
	(Man et al., 2012)
	middle temporal gyrus
	(Vandenberghe et al., 1996; Shinkareva et al., 2011; Fairhall and Caramazza, 2013; Devereux et al., 2013; Simanova et al., 2014; Handjaras et al., 2016)
	inferior temporal gyrus
	(Vandenberghe et al., 1996; Shinkareva et al., 2011; Fairhall and Caramazza, 2013; Simanova et al., 2014; Handjaras et al., 2016)
	fusiform gyrus
	(Vandenberghe et al., 1996; Moore and Price, 1999; Bright et al., 2004; Shinkareva et al., 2011; Fairhall and Caramazza, 2013; Simanova et al., 2014)
	hippocampus
	(Vandenberghe et al., 1996)
	parahippocampus
	(Bright et al., 2004; Fairhall and Caramazza, 2013; Handjaras et al., 2016)
	perirhinal cortex
	(Bright et al., 2004; Fairhall and Caramazza, 2013)
	superior parietal cortex
	(Shinkareva et al., 2011)
	inferior parietal cortex
	(Shinkareva et al., 2011; Handjaras et al., 2016)
	cingulate gyrus
	(Moore and Price, 1999; Fairhall and Caramazza, 2013; Handjaras et al., 2016)
	cuneus
	(Shinkareva et al., 2011)
	precuneus
	(Shinkareva et al., 2011; Handjaras et al., 2016; Fairhall and Caramazza, 2013; Popham et al., 2021)
	supramarginal gyrus
	(Shinkareva et al., 2011; Handjaras et al., 2016)
	angular gyrus
	(Shinkareva et al., 2011; Fairhall and Caramazza, 2013; Devereux et al., 2013; Simanova et al., 2014; Handjaras et al., 2016; Popham et al., 2021)
	intraparietal sulcus
	(Shinkareva et al., 2011; Devereux et al., 2013)
	temporoparietal junction
	(Vandenberghe et al., 1996; Handjaras et al., 2016)
	superior frontal gyrus
	(Fairhall and Caramazza, 2013)
	middle frontal gyrus
	(Fairhall and Caramazza, 2013; Handjaras et al., 2016)
	inferior frontal gyrus
	(Vandenberghe et al., 1996; Bright et al., 2004; Simanova et al., 2014; Liuzzi et al., 2017; Handjaras et al., 2016)
	precentral gyrus
	(Shinkareva et al., 2011)
	postcentral gyrus
	(Shinkareva et al., 2011)
	paracentral lobule
	(Shinkareva et al., 2011)
	supplementary motor area
	(Shinkareva et al., 2011)
right hemisphere	fusiform gyrus
	(Shinkareva et al., 2011; Simanova et al., 2014)
	superior temporal sulcus
	(Man et al., 2012)
	middle temporal gyrus
	(Handjaras et al., 2016)
	inferior temporal gyrus
	(Handjaras et al., 2016)
	angular gyrus
	(Handjaras et al., 2016; Popham et al., 2021)
	superior parietal cortex
	(Shinkareva et al., 2011)
	precuneus
	(Shinkareva et al., 2011; Popham et al., 2021; Fairhall and Caramazza, 2013; Handjaras et al., 2016)
	cingulate gyrus
	(Moore and Price, 1999; Fairhall and Caramazza, 2013; Handjaras et al., 2016; Jung et al., 2018)
	parahippocampus
	(Handjaras et al., 2016)
	inferior parietal cortex
	(Handjaras et al., 2016)
	paracentral lobule
	(Shinkareva et al., 2011)
	middle frontal gyrus
	(Simanova et al., 2014; Jung et al., 2018)
	superior frontal gyrus*
	(Jung et al., 2018)
	inferior frontal gyrus
	(Moore and Price, 1999; Simanova et al., 2014)

Appendix 3—table 1.

Candidates for modality-agnostic regions as identified by previous work.

All these regions were also found in our analysis, except for the 2 regions marked with an asterisk (*).

Appendix 4 Searchlight size

In order to optimize the size of the searchlight for this analysis, we first ran a searchlight analysis with a fixed radius of 10mm using a modality-agnostic decoder on the data from the first subject (sub-01). Due to the shape of the cortex this leads of searchlights that contain varying numbers of vertices (on average: 897.4; max: 1580; min: 399). By observing the decoding scores as function of the number of vertices we find that performance peaks at 750 vertices (cf. Table 1). The final searchlight analyses was therefore performed with a searchlight of a fixed number of 750 vertices. The average radius with this number of vertices was 9.41mm (max: 13.65mm).

# vertices	pairwise accuracy (mean)
250	51.81%
500	52.35%
750	55.96%
1000	55.49%
1250	54.60%
1500	52.96%

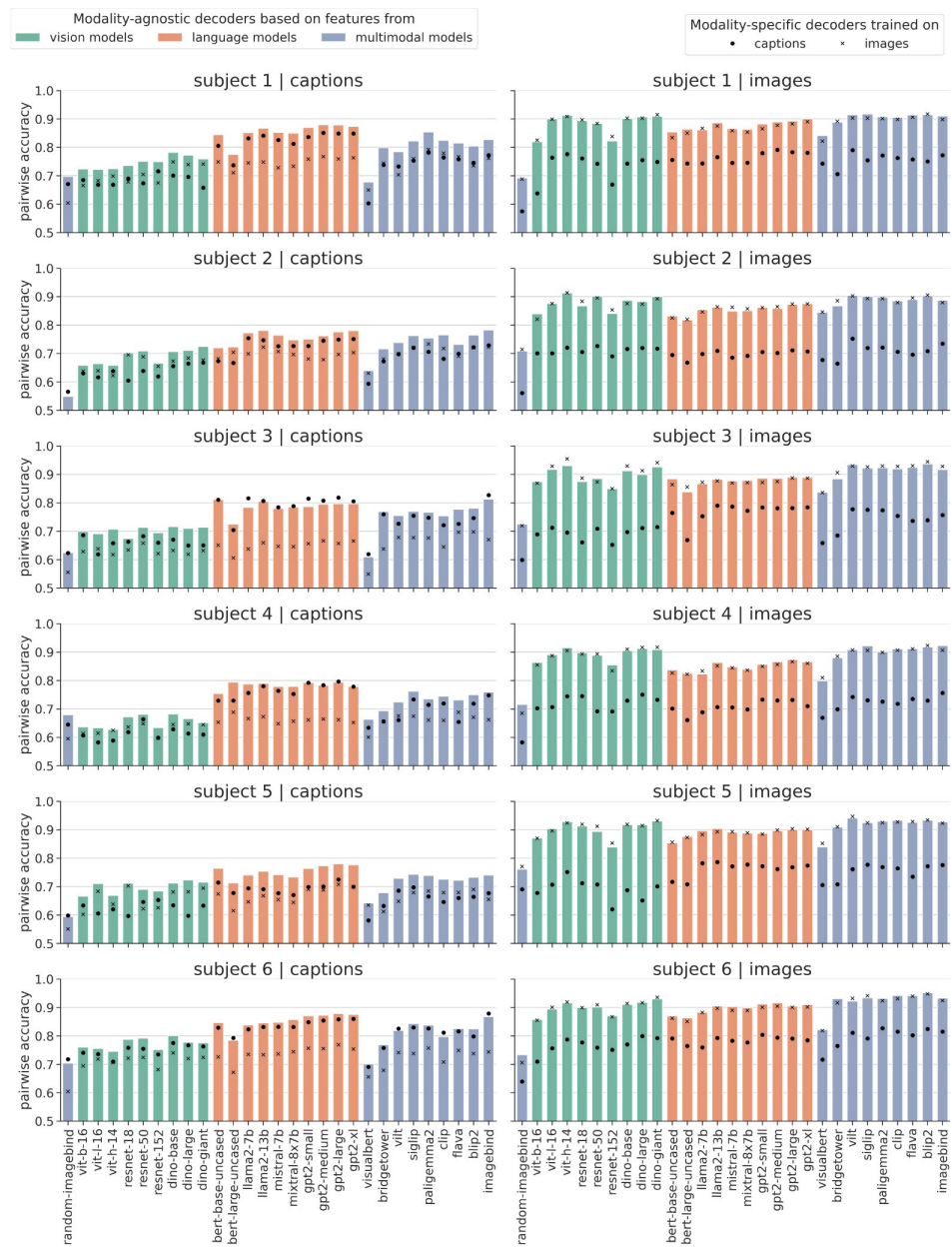
Appendix 4—table 1.

Average decoding scores (for images and captions) by number of vertices.

Scores calculated based on the results of a searchlight analysis with a radius of 10mm. The accuracy values were grouped into bins based on the number of vertices that the searchlight was based on.

Appendix 5 Per-subject results

Results for individual subjects can be found in **Figure 1** [↗](#). Among all subjects, we found similar converging results for decoding accuracies when comparing models, feature modalities, and modality-agnostic with modality-specific decoders.



Appendix 5—figure 1.

Pairwise accuracy per subject.

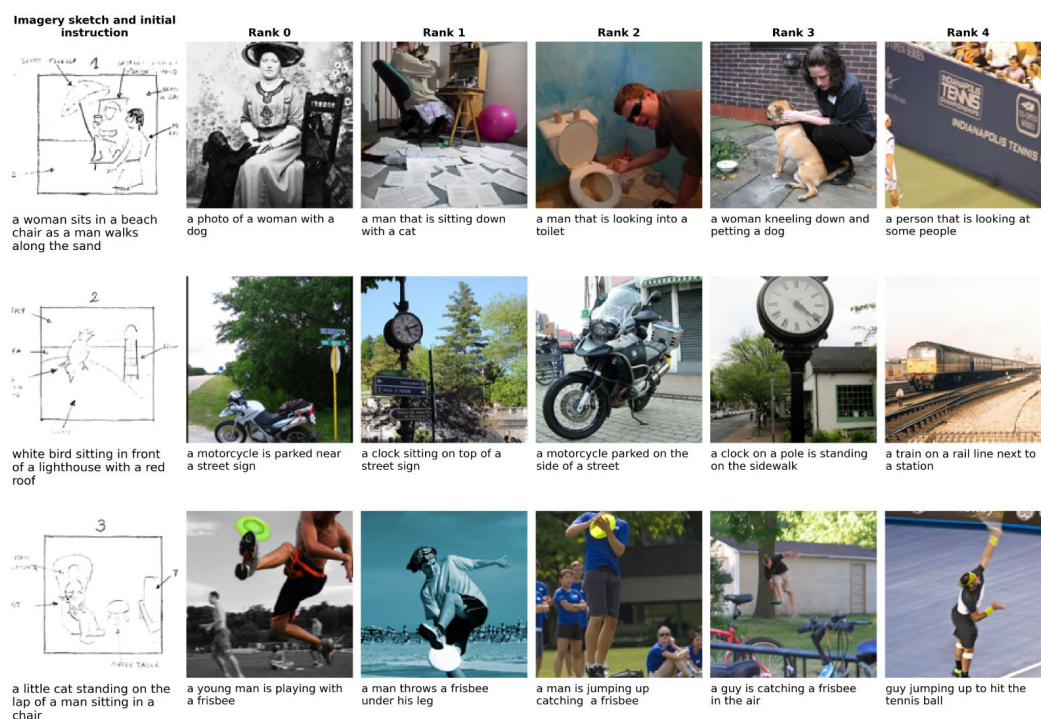
For details refer to [Figure 5](#)

Appendix 6 Qualitative Imagery Decoding Results

The following figures 1-6 present qualitative decoding results for the imagery conditions using a modality-agnostic decoder. We present the results separately for each subject as each subject chose an individual set of 3 stimuli to perform mental imagery on. In each plot, the leftmost column shows the caption that was used as initial instruction as well as the subject's sketch that they drew at the end of the experiment.

We used the same large candidate set of 41K stimuli as for the qualitative decoding results for the other conditions (see also Section Qualitative Decoding Results).

Overall, we found that the decoding quality for imagery stimuli lags behind that for trials with perceived stimuli. This was expected and confirms the quantitative results reported in Section Imagery Decoding. Still, in several cases some of the concepts are decoded correctly (e.g. a women in the first row of [Figure 1](#); winter sports in the second row of [Figure 3](#); laptops/screens and multiple people in the third row of [Figure 3](#)).



Appendix 6—figure 1.

Imagery decoding for Subject 1 using a modality-agnostic decoder.

The first column shows the caption that was used to stimulate imagery and on top of it the sketch of their mental image that the subjects draw at the end of the experiment. The 5 following columns show the candidate stimuli with highest similarity to the predicted features, in descending order. We display both the image and the caption of the candidate stimuli because the decoder is based on multimodal features that are extracted from both modalities. All images were taken from the CoCo dataset ([Lin et al., 2014](#)).

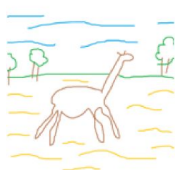
Imagery sketch and initial instruction



a woman sits in a beach chair as a man walks along the sand



a little cat standing on the lap of a man sitting in a chair



a lonely giraffe is walking in the middle of a grassy field

Rank 0



a couple of men sitting on some horses

Rank 1



a couple of men are playing a baseball game

Rank 2



a couple of men on a baseball field

Rank 3



a man standing leaning on a man sitting

Rank 4



a couple of people are sitting in a room



a tray sitting on a table full of food



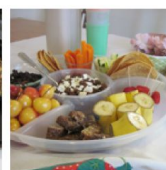
a tray covered in food on top of a table



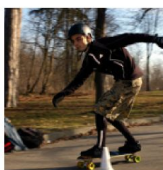
a small tray of food on a table



a bunch of food sitting on a tray



a tray of food sits on top of a table



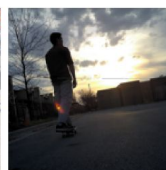
a man skateboarding on a road with pylons



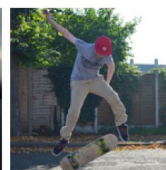
a guy jumping his skateboard off a curb



a man that is skateboarding down a road



a guy gliding down the road on a skateboard



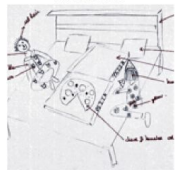
a boy jumps off his skateboard the street

Appendix 6—figure 2.

Imagery decoding for Subject 2 using a modality-agnostic decoder.

For further details refer to the caption of [Figure 1](#). All images were taken from the CoCo dataset (Lin et al., 2014).

Imagery sketch and initial instruction



a party of young people in a bedroom with a large box of pizza



a man walking across a field of snow holding skis and ski poles



two men are discussing something next to a desk with a laptop

Rank 0



a man and a dog are playing frisbee

Rank 1



a woman playing with a dog and frisbees

Rank 2



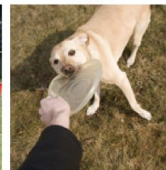
a young lady playing frisbee with her dog

Rank 3



a lady and a dog playing with a disc

Rank 4



a person wrestles over a frisbee with a dog



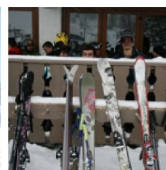
a snowboard being held upright in the snow



a snowboard standing up in a bank of snow



a person skiing on a mountain in mid air



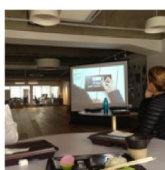
a row of skis sitting up against a fence



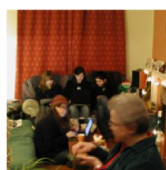
a plane flying in the sky during the day



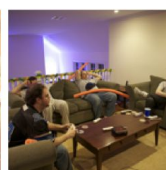
a group of people in a small room



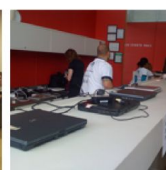
people watching a screen in a room



a bunch of people sit inside of a room



a group of people in a room by a table

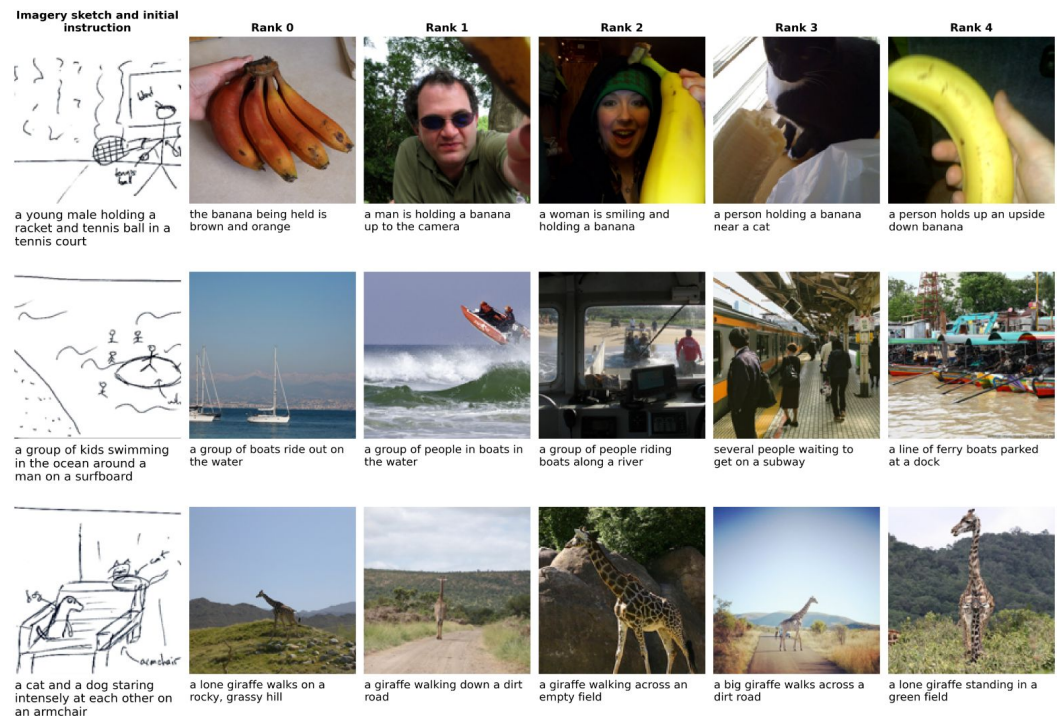


people stand near a desk with laptops on it

Appendix 6—figure 3.

Imagery decoding for Subject 3 using a modality-agnostic decoder.

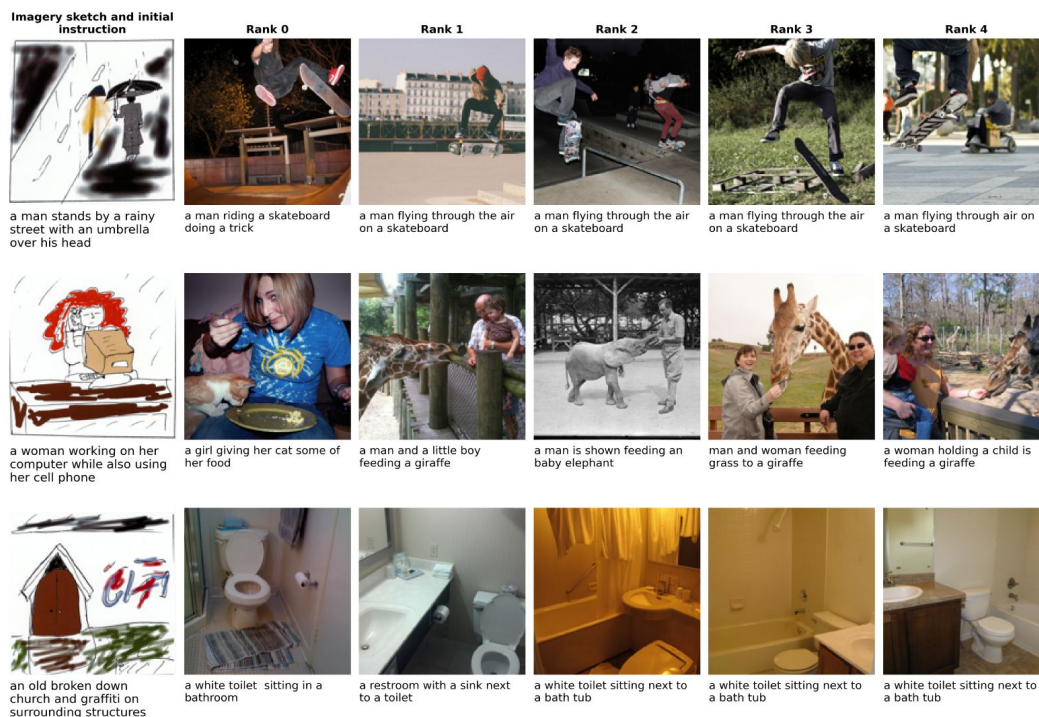
For further details refer to the caption of [Figure 1](#). All images were taken from the CoCo dataset (Lin et al., 2014).



Appendix 6—figure 4.

Imagery decoding for Subject 4 using a modality-agnostic decoder.

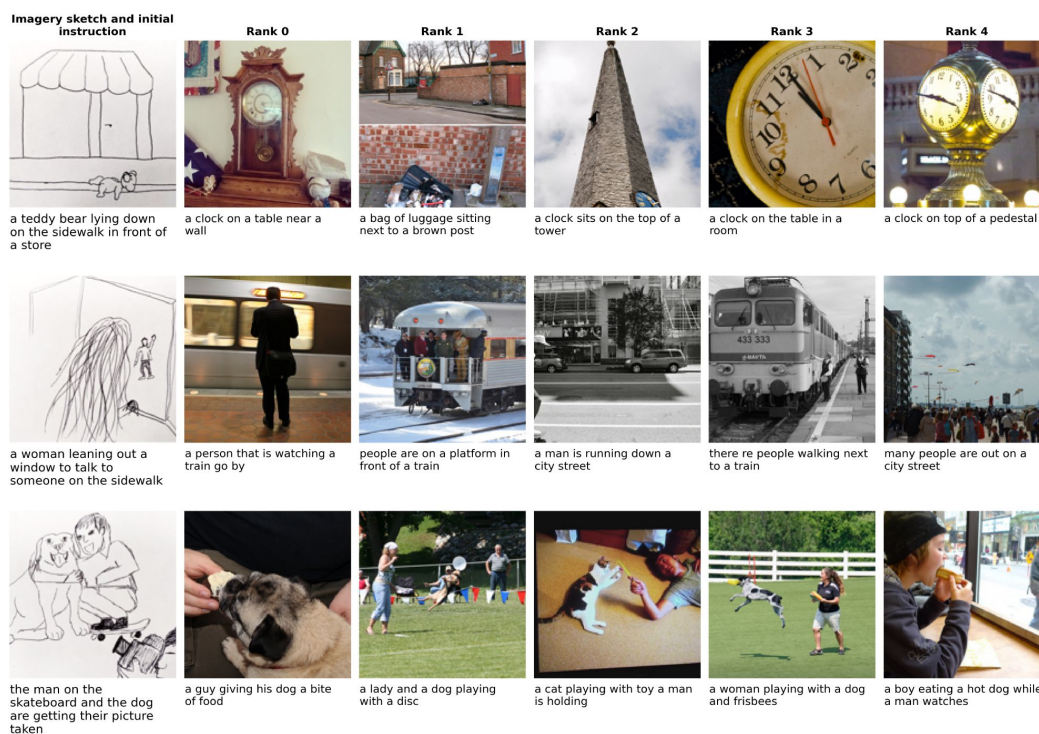
For further details refer to the caption of [Figure 1](#). All images were taken from the CoCo dataset (Lin et al., 2014).



Appendix 6—figure 5.

Imagery decoding for Subject 5 using a modality-agnostic decoder.

For further details refer to the caption of **Figure 1**. All images were taken from the CoCo dataset (Lin et al., 2014).



Appendix 6—figure 6.

Imagery decoding for Subject 6 using a modality-agnostic decoder. For further details refer to the caption of Figure 1.

All images were taken from the CoCo dataset (Lin et al., 2014 [↗](#)).

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Reviewer #1 (Public review):

Summary:

The authors introduce a densely-sampled dataset where 6 participants viewed images and sentence descriptions derived from the MS Coco database over the course of 10 scanning sessions. The authors further showcase how image and sentence decoders can be used to predict which images or descriptions were seen, using pairwise decoding across a set of 120 test images. The authors find decodable information widely distributed across the brain, with a left-lateralized focus. The results further showed that modality-agnostic models generally outperformed modality-specific models, and that data based on captions was not explained better by caption-based models but by modality-agnostic models. Finally, the authors decoded imagined scenes.

Strengths:

- (1) The dataset presents a potentially very valuable resource for investigating visual and semantic representations and their interplay.
- (2) The introduction and discussion are very well written in the context of trying to understand the nature of multimodal representations and present a comprehensive and very useful review of the current literature on the topic.

Weaknesses:

- (1) The paper is framed as presenting a dataset, yet most of it revolves around the presentation of findings in relation to what the authors call modality-agnostic representations, and in part around mental imagery. This makes it very difficult to assess the

manuscript, whether the authors have achieved their aims, and whether the results support the conclusions.

(2) While the authors have presented a potential use case for such a dataset, there is currently far too little detail regarding data quality metrics expected from the introduction of similar datasets, including the absence of head-motion estimates, quality of intersession alignment, or noise ceilings of all individuals.

(3) The exact methods and statistical analyses used are still opaque, making it hard for a reader to understand how the authors achieved their results. More detail in the manuscript would be helpful, specifically regarding the exact statistical procedures, what tests were performed across, or how data were pooled across participants.

(4) Many findings (e.g., Figure 6) are still qualitative but could be supported by quantitative measures.

(5) Results are significant in regions that typically lack responses to visual stimuli, indicating potential bias in the classifier. This is relevant for the interpretation of the findings. A classification approach less sensitive to outliers (e.g., 70-way classification) could avoid this issue. Given the extreme collinearity of the experimental design, regressors in close temporal proximity will be highly similar, which could lead to leakage effects.

(6) The manuscript currently lacks a limitations section, specifically regarding the design of the experiment. This involves the use of the overly homogenous dataset Coco, which invites overfitting, the mixing of sentence descriptions and visual images, which invites imagery of previously seen content, and the use of a 1-back task, which can lead to carry-over effects to the subsequent trial.

(7) I would urge the authors to clarify whether the primary aim is the introduction of a dataset and showing the use of it, or whether it is the set of results presented. This includes the title of this manuscript. While the decoding approach is very interesting and potentially very valuable, I believe that the results in the current form are rather descriptive, and I'm wondering what specifically they add beyond what is known from other related work. This includes imagery-related results. This is completely fine! It just highlights that a stronger framing as a dataset is probably advantageous for improving the significance of this work.

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Reviewer #2 (Public review):

Summary:

This study introduces SemReps-8K, a large multimodal fMRI dataset collected while subjects viewed natural images and matched captions, and performed mental imagery based on textual cues. The authors aim to train modality-agnostic decoders—models that can predict neural representations independently of the input modality - and use these models to identify brain regions containing modality-agnostic information. They find that such decoders perform comparably or better than modality-specific decoders and generalize to imagery trials.

Strengths:

(1) The dataset is a substantial and well-controlled contribution, with >8,000 image-caption trials per subject and careful matching of stimuli across modalities - an essential resource for testing theories of abstract and amodal representation.

(2) The authors systematically compare unimodal, multimodal, and cross-modal decoders using a wide range of deep learning models, demonstrating thoughtful experimental design and thorough benchmarking.

(3) Their decoding pipeline is rigorous, with informative performance metrics and whole-brain searchlight analyses, offering valuable insights into the cortical distribution of shared representations.

(4) Extension to mental imagery decoding is a strong addition, aligning with theoretical predictions about the overlap between perception and imagery.

Weaknesses:

While the decoding results are robust, several critical limitations prevent the current findings from conclusively demonstrating truly modality-agnostic representations:

(1) Shared decoding $\neq$ abstraction: Successful decoding across modalities does not necessarily imply abstraction or modality-agnostic coding. Participants may engage in modality-specific processes (e.g., visual imagery when reading, inner speech when viewing images) that produce overlapping neural patterns. The analyses do not clearly disambiguate shared representational structure from genuinely modality-independent representations. Furthermore, in Figure 5, the modality-agnostic encoder did not perform better than the modality-specific decoder trained on images (in decoding images), but outperformed the modality-specific decoder trained on captions (in decoding captions). This asymmetry contradicts the premise of a truly "modality-agnostic" encoder. Additionally, given the similar performance between modality-agnostic decoders based on multimodal versus unimodal features, it remains unclear why neural representations did not preferentially align with multimodal features if they were truly modality-independent.

(2) The current analysis cannot definitively conclude that the decoder itself is modality-agnostic, making "Qualitative Decoding Results" difficult to interpret in this context. This section currently provides illustrative examples, but lacks systematic quantitative analyses.

(3) The use of mental imagery as evidence for modality-agnostic decoding is problematic. Imagery involves subjective, variable experiences and likely draws on semantic and perceptual networks in flexible ways. Strong decoding in imagery trials could reflect semantic overlap or task strategies rather than evidence of abstraction.

The manuscript presents a methodologically sophisticated and timely investigation into shared neural representations across modalities. However, the current evidence does not clearly distinguish between shared semantics, overlapping unimodal processes, and true modality-independent representations. A more cautious interpretation is warranted. Nonetheless, the dataset and methodological framework represent a valuable resource for the field.

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Reviewer #3 (Public review):

Summary:

The authors recorded brain responses while participants viewed images and captions. The images and captions were taken from the COCO dataset, so each image has a corresponding caption, and each caption has a corresponding image. This enabled the authors to extract features from either the presented stimulus or the corresponding stimulus in the other modality. The authors trained linear decoders to take brain responses and predict stimulus

features. "Modality-specific" decoders were trained on brain responses to either images or captions, while "modality-agnostic" decoders were trained on brain responses to both stimulus modalities. The decoders were evaluated on brain responses while the participants viewed and imagined new stimuli, and prediction performance was quantified using pairwise accuracy. The authors reported the following results:

- (1) Decoders trained on brain responses to both images and captions can predict new brain responses to either modality.
- (2) Decoders trained on brain responses to both images and captions outperform decoders trained on brain responses to a single modality.
- (3) Many cortical regions represent the same concepts in vision and language.
- (4) Decoders trained on brain responses to both images and captions can decode brain responses to imagined scenes.

Strengths:

This is an interesting study that addresses important questions about modality-agnostic representations. Previous work has shown that decoders trained on brain responses to one modality can be used to decode brain responses to another modality. The authors build on these findings by collecting a new multimodal dataset and training decoders on brain responses to both modalities.

To my knowledge, SemReps-8K is the first dataset of brain responses to vision and language where each stimulus item has a corresponding stimulus item in the other modality. This means that brain responses to a stimulus item can be modeled using visual features of the image, linguistic features of the caption, or multimodal features derived from both the image and the caption. The authors also employed a multimodal one-back matching task, which forces the participants to activate modality-agnostic representations. Overall, SemReps-8K is a valuable resource that will help researchers answer more questions about modality-agnostic representations.

The analyses are also very comprehensive. The authors trained decoders on brain responses to images, captions, and both modalities, and they tested the decoders on brain responses to images, captions, and imagined scenes. They extracted stimulus features using a range of visual, linguistic, and multimodal models. The modeling framework appears rigorous, and the results offer new insights into the relationship between vision, language, and imagery. In particular, the authors found that decoders trained on brain responses to both images and captions were more effective at decoding brain responses to imagined scenes than decoders trained on brain responses to either modality in isolation. The authors also found that imagined scenes can be decoded from a broad network of cortical regions.

Weaknesses:

The characterization of "modality-agnostic" and "modality-specific" decoders seems a bit contradictory. There are three major choices when fitting a decoder: the modality of the training stimuli, the modality of the testing stimuli, and the model used to extract stimulus features. However, the authors characterize their decoders based on only the first choice—"modality-specific" decoders were trained on brain responses to either images or captions, while "modality-agnostic" decoders were trained on brain responses to both stimulus modalities. I think that this leads to some instances where the conclusions are inconsistent with the methods and results.

First, the authors suggest that "modality-specific decoders are not explicitly encouraged to pick up on modality-agnostic features during training" (line 137) while "modality-agnostic

decoders may be more likely to leverage representations that are modality-agnostic" (line 140). However, whether a decoder is required to learn modality-agnostic representations depends on both the training responses and the stimulus features. Consider the case where the stimuli are represented using linguistic features of the captions. When you train a "modality-specific" decoder on image responses, the decoder is forced to rely on modality-agnostic information that is shared between the image responses and the caption features. On the other hand, when you train a "modality-agnostic" decoder on both image responses and caption responses, the decoder has access to the modality-specific information that is shared by the caption responses and the caption features, so it is not explicitly required to learn modality-agnostic features. As a result, while the authors show that "modality-agnostic" decoders outperform "modality-specific" decoders in most conditions, I am not convinced that this is because they are forced to learn more modality-agnostic features.

Second, the authors claim that "modality-specific decoders can be applied only in the modality that they were trained on, while "modality-agnostic decoders can be applied to decode stimuli from multiple modalities, even without knowing a priori the modality the stimulus was presented in" (line 47). While "modality-agnostic" decoders do outperform "modality-specific" decoders in the cross-modality conditions, it is important to note that "modality-specific" decoders still perform better than expected by chance (figure 5). It is also important to note that knowing about the input modality still improves decoding performance even for "modality-agnostic" decoders, since it determines the optimal feature space—it is better to decode brain responses to images using decoders trained on image features, and it is better to decode brain responses to captions using decoders trained on caption features.

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